LIFE VAULT

(Smart Emergency & Digital Legacy System)

A PROJECT REPORT

Submitted by

DEEPTHI SHREE S

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THEJASWINI S

in partial fulfilment for the award of the course

AMC1361 – MOBILE APPLICATION DEVELOPMENT

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY



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Under the Guidance of

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DECLARATION BY THE CANDIDATES

We declare that, to the best of our knowledge, the work reported here has been composed solely by us and has not been submitted, either in whole or in part, for any previous application for a course.

Submitted for the project Viva-Voce held at K. Ramakrishnan College of Engineering on _____.

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ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF CSE

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

M1 To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.

M2 To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3 To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4 To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

PEO1 Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.

PEO2 Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.

PEO3 Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

PSO1 Expertise in tailoring ML algorithms and models to excel in designated applications and fields.

PSO2 Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

LifeVault is an Android-based personal emergency and digital legacy management application built using Kotlin, Jetpack Compose, and Firebase. It centralizes emergency medical information, documents, insurance details, and nominee access on one secure cloud platform with real-time synchronization. A single SOS tap instantly shares the user's live location by SMS with trusted contacts and nominees, while a web admin panel monitors users, SOS alerts, and feedback. The system also provides biometric/Google authentication, permission-based nominee access, and AI-assisted health guidance. Emergency profiles, medical records, important documents, insurance details, medicine reminders, digital legacy records and a password vault are managed in a single application, and a scannable QR emergency card exposes critical information when it is needed most. Nominees receive section-wise, read-only access and can follow an active SOS on their own dashboard.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
LifeVault is an Android-based personal emergency and digital legacy management application built using Kotlin, Jetpack Compose, and Firebase. It centralizes emergency medical information, documents, insurance details, and nominee access on one secure cloud platform with real-time synchronization. A single SOS tap instantly shares the user's live location by SMS with trusted contacts and nominees, while a web admin panel monitors users, SOS alerts, and feedback. The system also provides biometric/Googleauthentication, permission-based nominee access, and AI-assisted health guidance. Emergency profiles, medical records, important documents, insurance details, medicine reminders, digital legacy records and a password vault are managed in a single application, and a scannable QR emergency card exposes critical information when it is needed most. Nominees receive section-wise, read-only access and can follow an active SOS on their own dashboard.	PO1 - 2 PO2 - 3 PO3 - 2 PO4 - 3 PO5 - 3 PO6 - 3 PO7 - 2 PO8 - 3 PO9 - 3 PO10 -2 PO11 - 3 PO12 -3	PSO1 - 3 PSO2 - 3

Note: 1 – Low, 2 – Medium, 3 – High.

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LIST OF ABBREVIATIONS

S.NO	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	AIML	Artificial Intelligence and Machine Learning
3	API	Application Programming Interface
4	CO	Course Outcome
5	GPS	Global Positioning System
6	HTTPS	Hypertext Transfer Protocol Secure
7	mHealth	Mobile Health
8	MVVM	Model–View–ViewModel
9	NoSQL	Not Only SQL (document database)
10	PEO	Program Educational Objective
11	PIN	Personal Identification Number
12	PO	Program Outcome
13	PSO	Program Specific Outcome
14	QR	Quick Response (code)
15	RBAC	Role-Based Access Control
16	SDK	Software Development Kit
17	SMS	Short Message Service
18	SOS	Distress signal / emergency alert
19	UI	User Interface

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

In an emergency, critical information such as blood group, allergies, and current medications is often inaccessible at the moment it is needed most. Responders and family members rarely have immediate access to a person's medical history, important documents such as insurance papers and identity proofs are kept physically scattered, and the people closest to the victim are usually unaware of the victim's location.

Existing solutions depend on paper records or scattered applications, with no real-time synchronization and no controlled sharing with trusted people. LifeVault – Smart Emergency & Digital Legacy System is an Android-based application that unifies documents, medical records, nominee access, and instant live-location alerts in a single application. LifeVault is built using Kotlin and Jetpack Compose, follows MVVM with Clean Architecture, and uses Firebase Authentication, Cloud Firestore and Firebase Storage as its cloud backend. A web admin panel complements the mobile application by monitoring users, SOS alerts, and feedback in real time.

1.2 PROBLEM IDENTIFICATION

The following problems were identified during the study and form the basis of the LifeVault project:

- Responders and family members lack immediate access to a person's medical history during a crisis.
- Important documents (insurance, ID proofs) are physically scattered, making retrieval difficult.
- Nominees typically get either full access or none, with no granular permission control.
- Family members are not informed of the victim's location at the moment of an emergency.

1.3 OBJECTIVES

The primary objective of LifeVault is to provide a single, secure and always-available platform that prepares a user and the user's trusted people for emergencies. The specific objectives are:

- To centralize emergency profiles, medical records, and important documents on a secure cloud platform.
- To develop a granular, permission-based nominee access system for selective data sharing.
- To provide proactive safety features – SOS emergency response, medicine reminders, and AI-assisted health guidance.

Overall, the application aims to reduce the time taken to reach critical information in a crisis, to keep trusted contacts informed through live-location alerts, and to let users decide exactly which information each nominee may see.

1.4 PURPOSE AND IMPORTANCE

The purpose of LifeVault is to act as a dependable digital companion that keeps a person's emergency-critical information organised, secure and instantly shareable with the right people. Medical details, insurance policies, documents, medicine schedules and emergency contacts are normally stored in different places; LifeVault brings them together so that they can be retrieved quickly by the user, or – within the permissions granted – by a nominee.

The importance of the system lies in three areas. First, it improves emergency readiness through the emergency profile, QR emergency card and one-tap SOS with live location. Second, it replaces the all-or-nothing sharing model with section-wise nominee permissions, protecting privacy. Third, it connects digital legacy records (messages, accounts, letters and will records) with emergency preparedness, two areas that are usually handled separately.

1.5 DATA SOURCE DESCRIPTION

LifeVault does not rely on a pre-existing dataset. Its data originates from user-generated information and from device and cloud services:

- **User-entered data:** emergency profile (blood group, allergies, medical conditions, emergency contact), medical records, document details, insurance details, medicines, digital legacy records, password-vault entries, trusted contacts and nominee details.
- **Uploaded files:** important documents uploaded to Firebase Storage.
- **Device data:** the user's live GPS/network location, obtained through LocationManager when an SOS is triggered.
- **System-generated data:** an automatic activity log of user actions (profile update, nominee addition, SOS trigger), SOS records, and feedback (1–5 star rating with a written review).
- **AI responses:** health-guidance replies produced by the Gemini API for the AI Health Assistant.

The data is stored as NoSQL documents in Cloud Firestore. The usefulness of the emergency features depends on the completeness and accuracy of the information entered by the user.

1.6 PROJECT SUMMARIZATION

LifeVault is a mobile-based emergency and digital legacy solution that integrates secure data management, nominee access control, live SOS and location sharing, AI-assisted health guidance, and a web-based administration panel. The system is organised into nine modules: Authentication & Onboarding; Core Data Management; Emergency & Access Control; Live SOS & Location; Supporting Architecture; Intelligence & Notification; User & System; Security; and Web Admin Panel.

CHAPTER 2
LITERATURE SURVEY

2.1 PATIENT-CONTROLLED ACCESS TO MEDICAL INFORMATION

Mandl, K. D., Szolovits, P., and Kohane, I. S. (2001) examined how electronic medical records can be kept accessible and yet private. The paper argues for public standards and for giving patients control over who can see their records. Its central idea – that information should be available to authorised parties when needed while remaining under the patient's control – is directly reflected in LifeVault's nominee system, where the user decides which sections each nominee may view.

2.2 ROLE-BASED ACCESS CONTROL

Sandhu, R. S., Coyne, E. J., Feinstein, H. L., and Youman, C. E. (1996) presented a family of role-based access control (RBAC) models. The work shows how access rights can be tied to roles instead of being granted individually, simplifying administration and supporting the principle of least privilege. LifeVault applies the same principle in a lightweight form: the owner, nominee and administrator have different rights, and a nominee is limited to read-only access to the permitted data categories.

2.3 MOBILE HEALTH (mHEALTH) APPLICATIONS

Silva, B. M. C., Rodrigues, J. J. P. C., de la Torre Díez, I., López-Coronado, M., and Saleem, K. (2015) reviewed the state of mobile health in 2015. The review surveys the mHealth landscape and discusses opportunities as well as challenges, including the need to protect sensitive health data on mobile devices. This supports the decision to build LifeVault as a mobile application and to treat security of stored medical information as a core module.

2.4 MEDICATION-REMINDER APPLICATIONS

Armitage, L. C., Kassavou, A., and Sutton, S. (2020) carried out a systematic review of randomised controlled trials to assess whether mobile apps designed to support medication adherence are effective. The review reports promising but limited and varied evidence, which indicates that reminders are useful but are best combined with other supportive features. LifeVault's medicine reminder and AlarmManager-based notifications are therefore offered as part of a wider emergency-readiness platform rather than as a stand-alone tool.

2.5 CONVERSATIONAL AGENTS IN HEALTHCARE

Laranjo, L., Dunn, A. G., Tong, H. L., Kocaballi, A. B., Chen, J., Bashir, R., Surian, D., Gallego, B., Magrabi, F., Lau, A. Y. S., and Coiera, E. (2018) conducted a systematic review of conversational agents in healthcare. The review concludes that the evidence base was still limited and that most systems were at an early stage. This is relevant to LifeVault's AI Health Assistant, which provides medicine explanation and general health guidance through the Gemini API and should be positioned as informational support rather than as a replacement for professional medical advice.

2.6 DIGITAL LEGACY AND THE DIGITAL AFTERLIFE

Öhman, C., and Floridi, L. (2017) critically analysed the "digital afterlife industry" and the ethical issues surrounding the data a person leaves behind. The paper highlights that digital remains need deliberate and respectful handling. This motivates LifeVault's Digital Legacy feature, where messages, accounts, letters and will records are stored securely and tied to the same nominee permission model.

2.7 SUMMARY OF THE LITERATURE

Table 2.1 Summary of Reviewed Literature

S. No	Authors	Year	Research Focus	Key Findings	Relevance to LifeVault
1	Mandl, Szolovits & Kohane	2001	Accessible yet private electronic medical records	Patients should control access; public standards needed	Nominee permission model

S. No	Authors	Year	Research Focus	Key Findings	Relevance to LifeVault
2	Sandhu, Coyne, Feinstein & Youman	1996	Role-based access control models	Roles simplify administration and enable least privilege	Owner / nominee / admin rights
3	Silva, Rodrigues, de la Torre Díez, López-Coronado & Saleem	2015	Review of mobile health	Mobile health offers opportunities but raises data-protection challenges	Mobile-first design; Security Module
4	Armitage, Kassavou & Sutton	2020	Efficacy of medication-adherence apps	Evidence promising but limited and varied	Medicine Reminder and scheduler
5	Laranjo et al.	2018	Conversational agents in healthcare	Evidence base limited; systems at an early stage	AI Health Assistant (informational use)
6	Öhman & Floridi	2017	Digital afterlife industry and ethics	Digital remains need careful, ethical handling	Digital Legacy feature

2.8 RESEARCH GAP

The literature and the study of existing solutions show that individual aspects of the problem are addressed separately, while an integrated emergency-ready solution is missing. The gaps identified are:

- Information exists, but emergency-ready information is missing.
- Documents can be stored, but medical, insurance and emergency data are not unified.
- Access can be shared, but not selectively based on nominee permissions.
- SOS can send an alert, but continuous family/nominee awareness is limited.
- Digital legacy exists separately, but is rarely connected with emergency preparedness.

LifeVault bridges these gaps by bringing everything together in one connected platform.

CHAPTER 3
PROJECT METHODOLOGY

3.1 PROPOSED WORKFLOW

LifeVault is built on MVVM with Clean Architecture. The Presentation layer is implemented with Jetpack Compose, the Domain layer holds the business logic, and the Data layer uses a Firebase repository. Hilt provides dependency injection and Navigation Compose manages screen navigation.

The workflow begins with the splash screen and three-page onboarding, after which the user registers or logs in. Firebase Authentication verifies identity using Email/Password, Google Sign-In or Biometric login. Once authenticated, the user enters emergency, medical, document, insurance, medicine, legacy and password-vault data, which Cloud Firestore stores and synchronises in real time. Uploaded files are held in Firebase Storage.

Nominees use a separate credential system. Before any data is displayed, the stored permissions of the nominee are checked, so that only the sections granted by the user are exposed.

The SOS flow is: SOS tap → live location fix → SMS with a map link to trusted contacts and nominees → live location updated in Firestore → visible to nominees on the Nominee Dashboard and to administrators on the Live SOS Monitor. The user ends the flow with “I’m Safe”. Every user action is logged automatically to form a real-time activity timeline.

3.2 ARCHITECTURE DIAGRAM

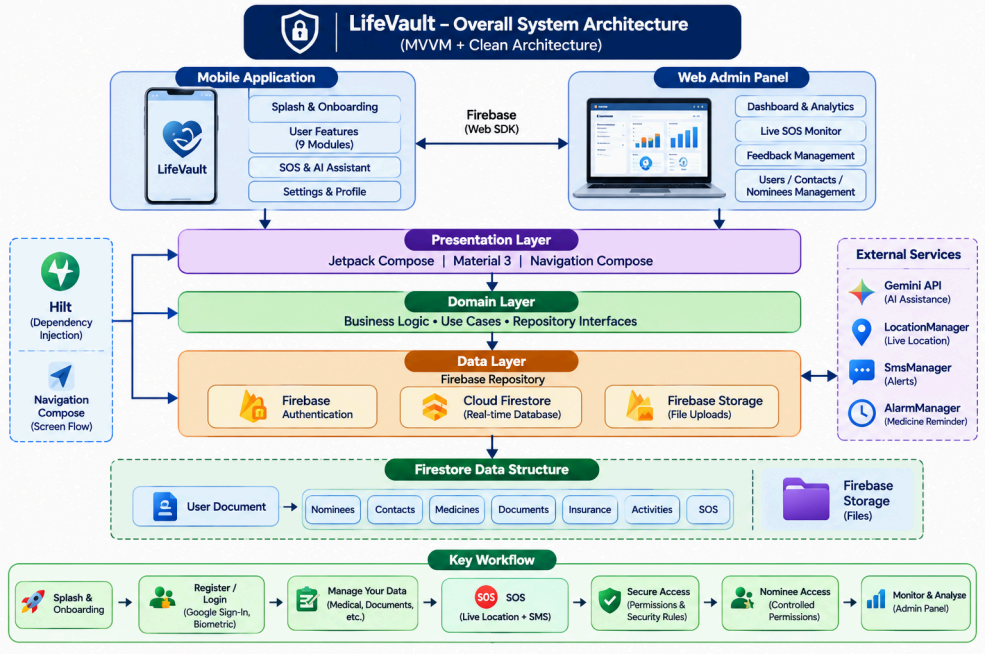


Figure 3.2.1 Architecture Diagram

The diagram represents the overall architecture of LifeVault using MVVM and Clean Architecture. It shows how the mobile application and Web Admin Panel interact through the Presentation, Domain, and Data layers. Jetpack Compose manages the user interface, while the Domain layer handles business logic and the Data layer connects with Firebase Authentication, Cloud Firestore, and Firebase Storage. Hilt manages dependency injection, and Navigation Compose controls screen navigation. External services such as Gemini API, LocationManager, SmsManager, and AlarmManager support AI assistance, live-location sharing, emergency alerts, and medicine reminders. Firestore stores user information and related records, while Firebase Storage maintains uploaded files. The architecture also illustrates the secure data flow between users, nominees, and administrators through permission-based access and centralized monitoring.

3.3 TECHNOLOGIES USED

Table 3.1 Technologies Used

Layer	Technology
Language	Kotlin
UI	Jetpack Compose, Material 3
Architecture	MVVM + Clean Architecture
Dependency Injection	Hilt
Navigation	Navigation Compose
Backend	Firebase Authentication, Cloud Firestore, Firebase Storage
AI	Gemini API
Location & Alerts	LocationManager, SmsManager, AlarmManager
Web	HTML, CSS, JavaScript, Firebase Web SDK, Netlify

3.4 DATA STRUCTURE ANALYSIS

Data is stored in a NoSQL document structure. Each user's root document holds the subcollections nominees, trusted contacts, medicines, documents, insurance, activities, and sos. Every user action (profile update, nominee addition, SOS trigger) is logged automatically, forming a real-time activity timeline.

Table 3.2 Firestore Data Organisation

Collection / Subcollection	Purpose (from project description)
User root document	Root document of each user; holds the subcollections below
nominees	Nominee details with permission arrays
trusted contacts	Contacts who receive emergency SMS
medicines	Medicine reminders (dose, frequency, status)
documents	Document details (files in Firebase Storage)
insurance	Policy details, validity, claim information
activities	Automatic activity timeline
sos	SOS records and live location
feedback (top-level)	Star ratings and reviews for admin analysis

CHAPTER 4
RELEVANCE OF THE PROJECT

4.1 ADDRESSING USER NEEDS

LifeVault addresses the needs identified in Section 1.2. The emergency profile and QR emergency card make blood group, allergies, medical conditions and emergency contact information available at the moment of need. Documents, insurance details and medical records are held in one place instead of being scattered physically. The nominee module replaces all-or-nothing sharing with section-wise permissions, and the SOS module informs trusted contacts and nominees of the user's live location.

4.2 ENHANCEMENT TO EMERGENCY PREPAREDNESS

The application enhances preparedness by combining proactive and reactive features. Proactively, the user keeps records current, sets medicine reminders and records digital legacy information. Reactively, one tap of the SOS button shares live location, with updates in Firestore every 10 seconds and a fresh SMS every 2 minutes. The activity timeline and the web admin panel add monitoring, allowing SOS alerts and feedback to be reviewed in real time.

4.3 BENEFITS AND LIMITATIONS

Benefits:

- All emergency-relevant information is centralised on a secure cloud platform with real-time synchronisation.
- Granular, permission-based and read-only nominee access protects privacy.
- One-tap SOS with live-location sharing keeps family and nominees informed.
- Email/Password, Google and Biometric authentication together with PIN app lock protect the application.
- AI-assisted health guidance and medicine reminders support day-to-day health management.

Limitations:

- The application is currently available for Android only.
- Background live tracking, push notifications for nominees and automatic fall detection are not yet available (listed under Future Scope).
- SOS delivery depends on the user granting location and SMS permissions, and on device and network conditions. The AI Health Assistant provides general guidance only and does not replace professional medical advice.

CHAPTER 5
MODULE DESCRIPTION

5.1 AUTHENTICATION & ONBOARDING MODULE

This module is the entry point of LifeVault. It introduces new users to the application and verifies the identity of every person who uses it, whether an account owner or a nominee. Because the application stores sensitive medical and personal information, no data screen is reachable until authentication succeeds.

When the application starts, a branded animated splash screen is displayed, followed by a three-page onboarding introduction for new users. The user then registers or logs in. Three sign-in methods are supported: Email/Password, Google Sign-In and Biometric login. Identity verification is performed by Firebase Authentication.

If a password is forgotten, the user requests a Firebase email reset. The screen provides a resend timer and a shortcut to open the email application, and a notification confirms the action. Nominees do not use the owner's account: a separate credential-based Nominee Login is provided, which leads to the permission-restricted Nominee Dashboard described in Module 3.

Key Functions

- **Splash Screen** – branded animated entry screen.
- **Onboarding** – three-page introduction for new users.
- **Register / Login** – Email-Password, Google Sign-In and Biometric Login.
- **Forgot Password** – Firebase email reset with resend timer and email-app shortcut.
- **Nominee Login** – separate credential-based login for nominees.

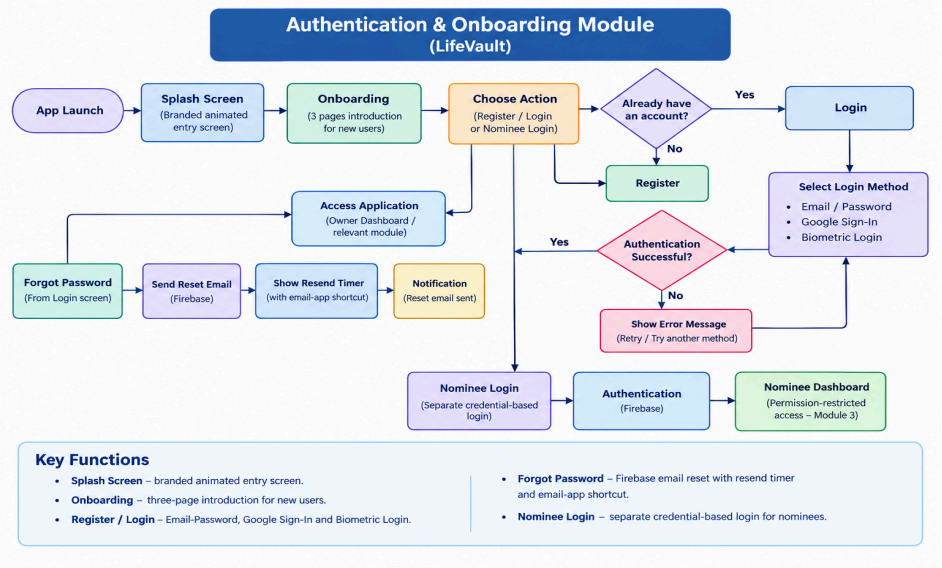


Figure 5.1.1 Authentication & Onboarding Module Diagram

5.2 CORE DATA MANAGEMENT MODULE

This module is the information store of LifeVault and holds everything a person or a nominee may need during an emergency. All records are saved to Cloud Firestore under the user's own document, so that they are synchronised in real time and can be recovered on sign-in.

The Dashboard gives real-time statistics and quick access to every feature. The Emergency Profile stores blood group, allergies, medical conditions and the emergency contact. Medical Records keep doctor, hospital, conditions, medications and vaccination details. The Documents feature stores document details, with the files themselves uploaded to Firebase Storage. The Insurance feature stores policy details, validity and claim information.

Further features in this module are the Medicine Reminder (dose, frequency and taken/upcoming status), Digital Legacy (messages, accounts, letters and will records), the Password Vault (secure storage of website and application login details) and the QR Emergency Card, a scannable card that contains critical emergency information. The Storage Summary and Secure Vault screens present a summary of what has been stored.

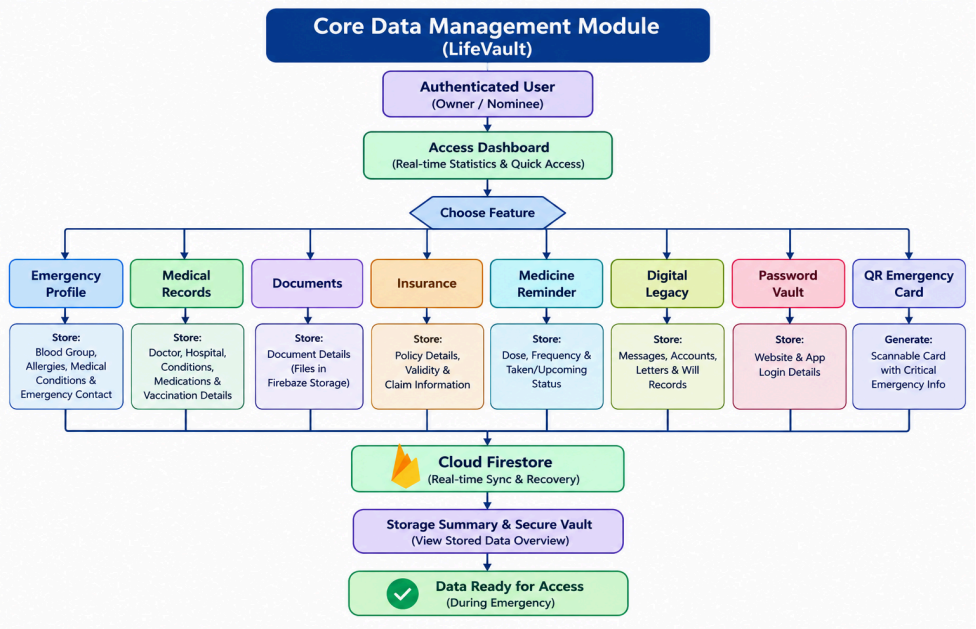


Figure 5.2.1 Core Data Management Module Diagram

5.3 EMERGENCY & ACCESS CONTROL MODULE

This module decides who can see the user's information and who is alerted in an emergency. It replaces the all-or-nothing access model with section-wise permissions.

In Nominee Management the user adds a nominee and selects the sections that nominee may view. The permissions are stored as arrays and checked every time the nominee signs in. The Nominee Dashboard therefore shows only the granted sections, in read-only mode, together with an SOS alert card whenever the user has an active SOS.

Trusted Contacts are the people who receive the emergency location through SMS. The SOS Module offers one-tap live-location alerts together with the options I'm Safe, Police (100) and Ambulance (108). The Secure Vault screen provides a real-time summary of the stored information.

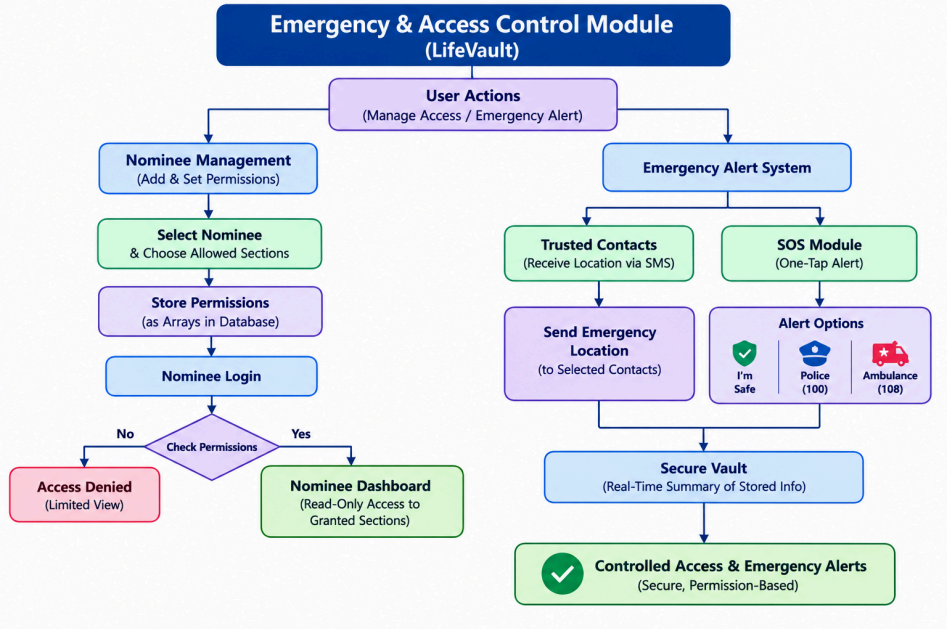


Figure 5.3.1 Emergency & Access Control Module Diagram

5.4 LIVE SOS & LOCATION MODULE

This module is the heart of the emergency response. Its purpose is to let a person in danger alert trusted people with one tap and to keep them informed of the person's position until the person is safe.

The first time SOS is used, the application requests Location and SMS permissions; they are requested only at that point. After permission is granted, the application obtains the user's live GPS/network location, and sends a Google Maps location link through SMS to the trusted contacts. While the SOS is active, the live location in Firestore is updated every 10 seconds and a fresh SMS is sent every 2 minutes.

The active SOS is shown on the Nominee Dashboard and on the administrator's Live SOS Monitor. When the user taps "I'm Safe", a safe message is sent and the feedback/rating prompt opens, ending the SOS session.

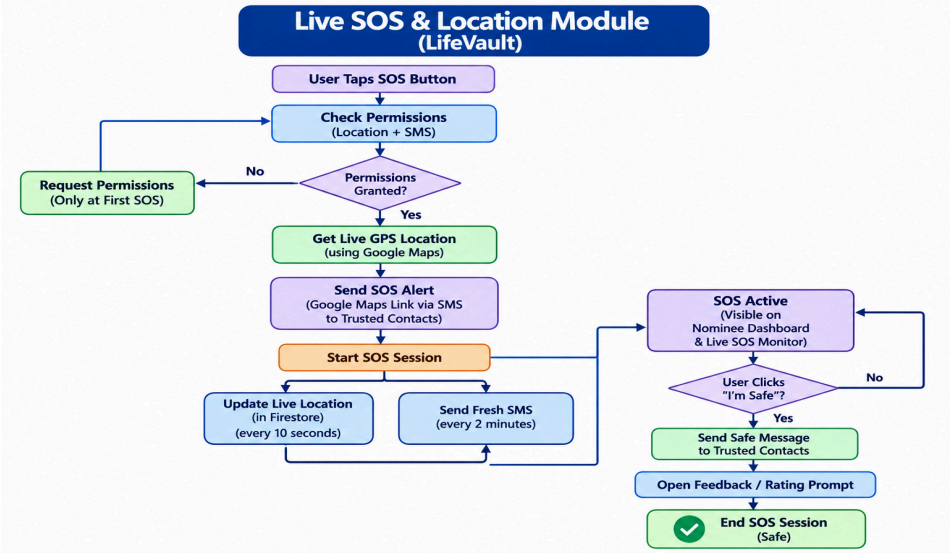


Figure 5.4.1 Live Sos & Location Module Diagram

5.5 SUPPORTING ARCHITECTURE MODULE

This module contains the technical foundation used by every other module. It does not appear as a separate screen but determines how the application is structured, so that features can be added without disturbing existing ones.

Navigation Compose centralises navigation between screens. The Theme module defines Material 3 colours, typography and shapes. Hilt supplies dependencies throughout the application. The Repository module is an abstraction layer over Firebase, so that screens and business logic do not call Firebase directly. Common UI Components provide reusable buttons, text fields, dialogs, toasts and helpers, giving a consistent look and behaviour. The SOS Manager handles location updates, SMS delivery and live synchronisation.

Together these components implement the MVVM + Clean Architecture design: Presentation (Jetpack Compose), Domain (business logic) and Data (Firebase repository) layers.

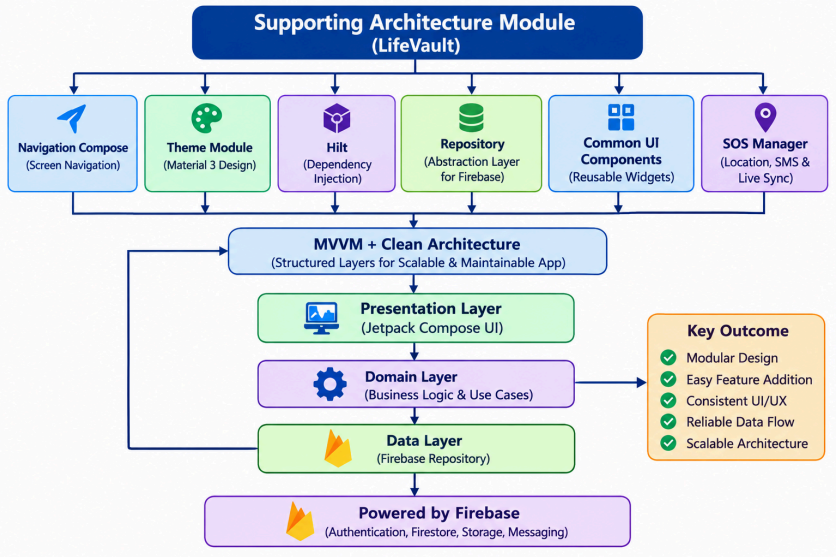


Figure 5.5.1 Supporting Architecture Module Diagram

5.6 INTELLIGENCE & NOTIFICATION MODULE

This module adds proactive assistance on top of stored data. It has three parts: the AI Health Assistant, Alerts, and the Medicine Notification Scheduler.The AI Health Assistant is a chat feature powered by the Gemini API. The user can ask for an explanation of a medicine or for general health guidance, and the reply appears in the chat. It is meant for information and not as a substitute for a doctor.

The Alerts screen is a consolidated view of medicine reminders and recent activity. The Medicine Notification Scheduler uses AlarmManager to schedule notifications for the reminders.

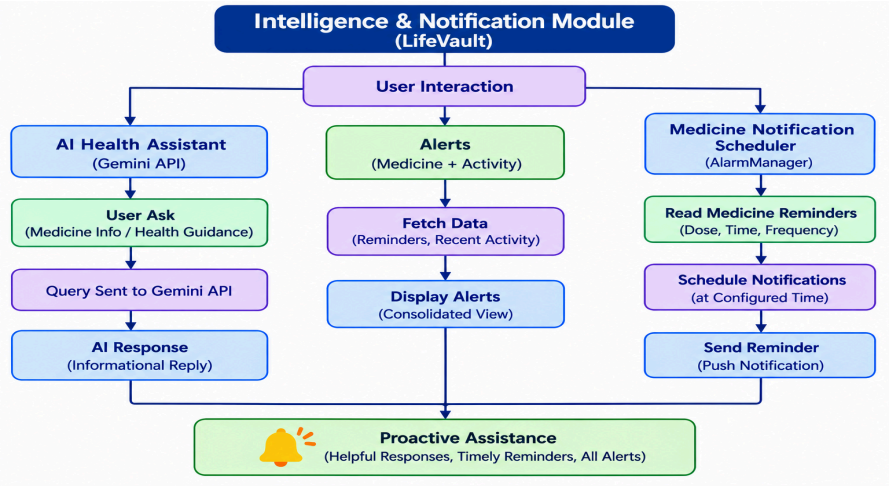


Figure 5.6.1 Intelligence & Notification Module Diagram

5.7 USER & SYSTEM MODULE

This module manages everything about the user's own account and the application's settings. The Profile screen shows the user's information and provides logout. Edit Profile updates personal information, and Recent Activity displays the activity history generated automatically from user actions.

The Settings screen provides the theme (dark/light), PIN Lock, Help, About, Feedback, Privacy Policy and Terms. A User Manual page is available for guidance. The Feedback feature lets users give a 1–5 star rating with a written review, which is stored in Firestore and later reviewed by administrators in the web panel.

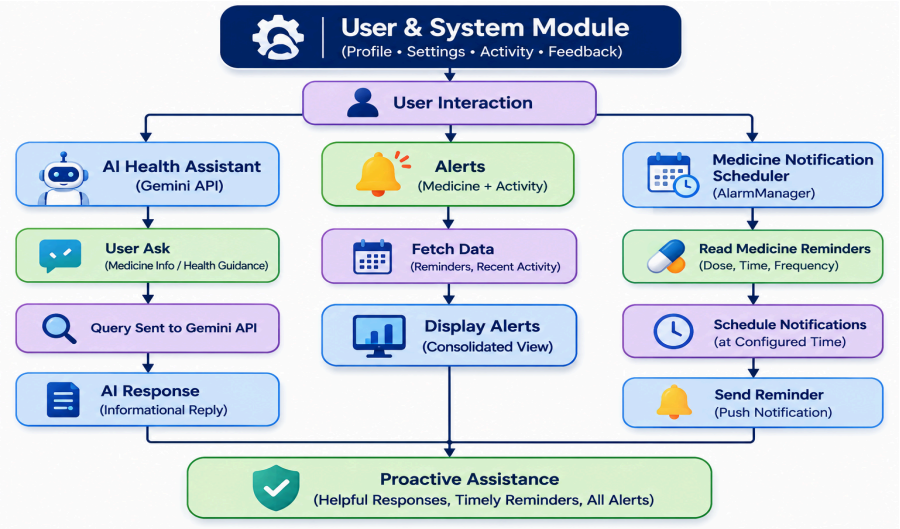


Figure 5.7.1 User & System Module Diagram

5.8 SECURITY MODULE

Because LifeVault stores medical, financial and personal information, security is applied at several levels. Authentication Security uses Email/Password, Google Sign-In and Biometric/PIN app lock. Firestore Security Rules control access to personal and admin data. Permission-Based Access ensures that nominees can view only the sections granted to them. Runtime Permissions for location and SMS are requested only during SOS. Secure Communication: Firebase communication uses HTTPS.These layers complement each other: even if a nominee is signed in, the data rules and permission arrays restrict what is shown; even if the device is unlocked, the PIN/biometric app lock guards the application.

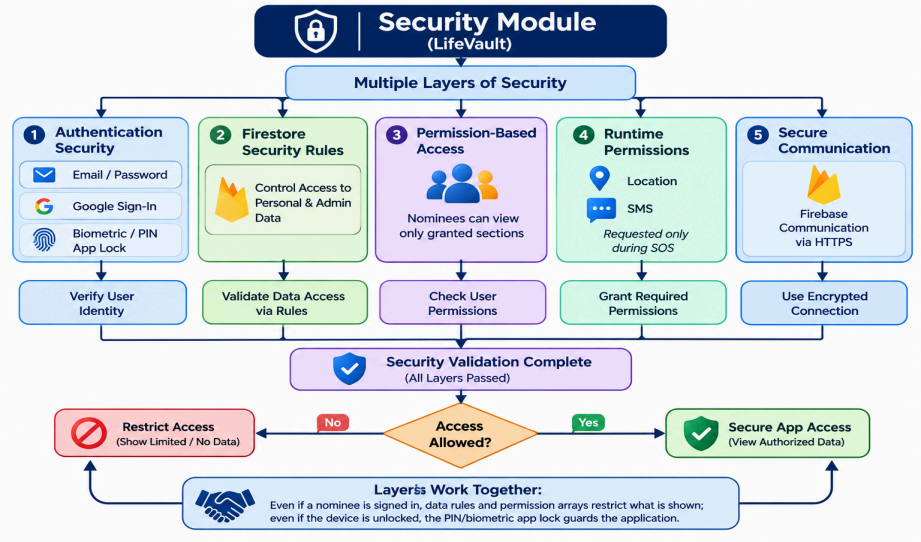


Figure 5.8.1 Security Module Diagram

5.9 WEB ADMIN PANEL MODULE

The Web Admin Panel allows authorised administrators to monitor the system. It is a web application built with HTML, CSS, JavaScript and the Firebase Web SDK and hosted on Netlify. A Landing Website with an animated promotional page presents features, team and details. After an Admin Login restricted to authorised administrators, the Dashboard shows total users, active SOS, SOS alerts and average rating, and Analytics provides a rating chart and a 7-day SOS chart.

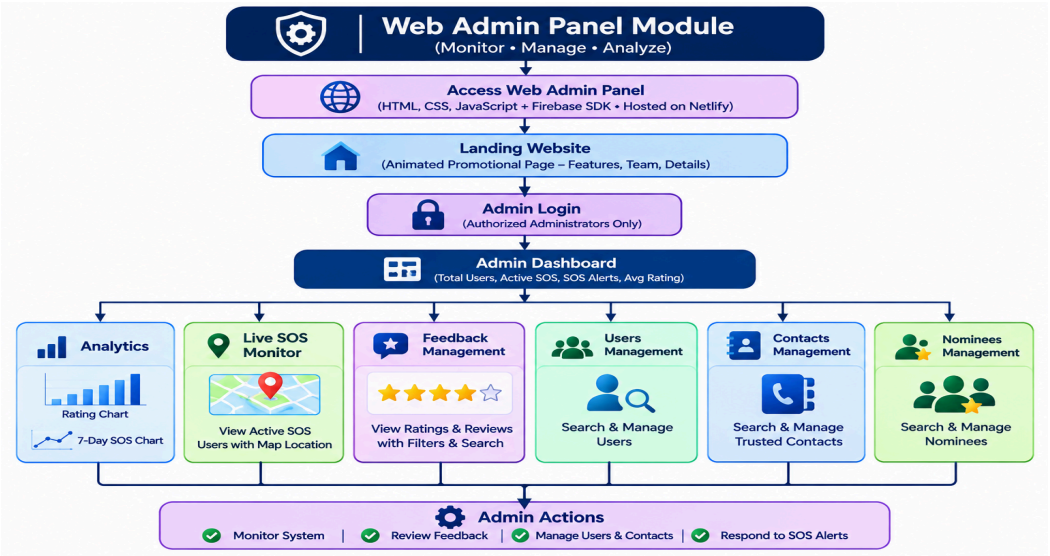


Figure 5.9.1 Web Admin Panel Module Diagram

CHAPTER 6
RESULTS AND DISCUSSION

6.1 RESULTS

The LifeVault mobile application and web-based Admin Panel have been successfully developed, and the expected project functionalities are reflected in the completed system. The application provides a centralized platform for managing users' important medical, personal, financial, and digital legacy information in a secure and organized manner. It offers a user-friendly interface through which users can access various features and manage their essential information conveniently.The application includes Splash and Onboarding Screens that introduce the purpose and features of LifeVault. The Registration and Login Screens support account creation and secure authentication through Email/Password and Google Sign-In.

The Nominee Management feature allows users to create nominees and provide them with access to selected information. Nominees can log in and view the sections authorized by the primary user through their dedicated dashboard. The Storage Summary provides an overview of stored information. These features ensure that important information can be made available to trusted individuals according to the permissions granted by the user.

Overall, the completed LifeVault application brings together medical information management, personal document storage, emergency assistance, digital legacy management, nominee access, AI-based health guidance, and administrative monitoring within a single platform. The developed system provides a convenient and organized approach to managing essential life-related information while maintaining controlled access and security.

6.2 DISCUSSION

The design of LifeVault responds directly to the research gaps identified in Section 2.8. Table 6.2 relates each gap to the feature that addresses it. The table describes design intent; whether each feature performs as intended is established only by the test results in Chapter 7.

Table 6.1 Research Gaps and Corresponding LifeVault Features

Research gap	LifeVault feature
Emergency-ready information is missing	Emergency Profile, QR Emergency Card
Medical + insurance + emergency data not unified	Core Data Management Module on one cloud platform
Access not shared selectively	Section-wise nominee permissions with read-only Nominee Dashboard
Limited continuous family/nominee awareness during SOS	Live location in Firestore every 10 s, SMS every 2 min, Nominee Dashboard SOS card
Digital legacy not connected with emergency preparedness	Digital Legacy records inside the same application and permission model

The application is Android-only, relies on user-entered data, and uses a Firebase backend. The AI Health Assistant gives general guidance. These factors, together with the items listed under Future Scope, define the current boundaries of the system.

CHAPTER 7
TESTING, VALIDATION AND QUALITY ASSURANCE

7.1 INTRODUCTION

Testing confirms that LifeVault behaves as specified, validation confirms that it meets the objectives in Section 1.3, and quality assurance confirms that the work follows good engineering and security practice. This chapter defines the test plan, the test cases with their expected results, and the fields in which the actual results and status must be recorded.

7.2 TESTING STRATEGY

- **Functional testing** of every feature of the nine modules.
- **Integration testing** of the app with Firebase Authentication, Cloud Firestore, Firebase Storage, Gemini API, LocationManager, SmsManager and AlarmManager.
- **Security and permission testing** of authentication, Firestore rules, nominee permissions and runtime permissions.
- **UI/UX testing** of navigation, theme and consistency across screens.
- **End-to-end testing** of the SOS scenario involving the owner, nominee and administrator.
- **Compatibility testing** on different Android versions and screen sizes.

7.3 TEST ENVIRONMENT

Table 7.1 Test Environment

Item	Details
Device model(s)	**Samsung Galaxy / Redmi / Vivo (actual device model)**
Android version(s)	Android 12 / 13 / 14 (actual version)
App version / build	LifeVault v1.0

Item	Details
Network type(s)	Wi-Fi and Mobile Data
Firebase project used	Firebase (Authentication, Cloud Firestore, Firebase Storage)
Admin panel URL / browser	Netlify Hosted URL / Google Chrome

7.4 TEST CASES

7.4.1 Authentication & Onboarding

Table 7.2 Test Cases – Authentication & Onboarding

TC ID	Test Case / Steps	Expected Result	Status
TC-AUTH-01	Launch the app for the first time	Animated splash is shown, then the three-page onboarding	**Pass**
TC-AUTH-02	Swipe through onboarding pages	All three introduction pages are shown and the user can proceed to register/login	**Pass**
TC-AUTH-03	Register with valid details	Account is created through Firebase Authentication	**Pass**
TC-AUTH-04	Login with valid email and password	User is authenticated and reaches the Dashboard	**Pass**
TC-AUTH-05	Login with wrong password	Error message shown; access is denied	**Pass**
TC-AUTH-06	Google Sign-In	User is authenticated with the Google account	**Pass**
TC-AUTH-07	Biometric login on a device with enrolled biometrics	Access is granted after biometric verification	**Pass**
TC-AUTH-08	Forgot password: request reset email	Reset email is sent; resend timer and email-app shortcut are shown	**Pass**
TC-AUTH-09	Nominee login with valid credentials	Nominee Dashboard opens	**Pass**
TC-AUTH-10	Nominee login with invalid credentials	Login is rejected	**Pass**

7.4.2 Core Data Management

Table 7.3 Test Cases – Core Data Management

TC ID	Test Case / Steps	Expected Result	Status
TC-DATA-01	Open Dashboard after login	Real-time statistics and quick-access items are displayed	**Pass**
TC-DATA-02	Enter emergency profile (blood group, allergies, conditions, emergency contact) and save	Data is saved and displayed correctly on the profile page	**Pass**

TC ID	Test Case / Steps	Expected Result	Status
TC-DATA-03	Add a medical record (doctor, hospital, conditions, medications, vaccination)	Record is saved and listed	**Pass**
TC-DATA-04	Upload an important document	File is uploaded to Firebase Storage and its details are listed	**Pass**
TC-DATA-05	Add insurance policy details	Policy details, validity and claim information are saved and shown	**Pass**
TC-DATA-06	Add a medicine reminder (dose, frequency)	Reminder is saved with taken/upcoming status	**Pass**
TC-DATA-07	Add a digital legacy record	Record is saved under Digital Legacy	**Pass**
TC-DATA-08	Add a password-vault entry	Entry is saved and shown in the vault list	**Pass**
TC-DATA-09	Open the QR Emergency Card	A scannable QR card with critical emergency information is displayed	**Pass**
TC-DATA-10	Close and reopen the app	Previously saved data is retrieved from the cloud	**Pass**

7.4.3 Emergency & Access Control

Table 7.4 Test Cases – Emergency & Access Control

TC ID	Test Case / Steps	Expected Result	Status
TC-ACC-01	Create a nominee and select permitted sections	Nominee is created with the selected permissions	**Pass**
TC-ACC-02	Login as nominee and open the Nominee Dashboard	Only the permitted sections are visible, in read-only mode	**Pass**
TC-ACC-03	Nominee tries to reach a section that was not granted	Section is not accessible	**Pass**
TC-ACC-04	Add a trusted contact	Contact is saved and used as an SOS recipient	**Pass**
TC-ACC-05	Open the SOS screen	I'm Safe, Police (100) and Ambulance (108) options are shown	**Pass**
TC-ACC-06	Select Police / Ambulance options	The corresponding emergency action starts	**Pass**
TC-ACC-07	Active SOS while nominee dashboard is open	SOS alert card appears on the Nominee Dashboard	**Pass**
TC-ACC-08	Open Secure Vault / Storage Summary	Real-time summary of stored information is shown	**Pass**

7.4.4 Live SOS & Location

Table 7.5 Test Cases – Live SOS & Location

TC ID	Test Case / Steps	Expected Result	Status
TC-SOS-01	Use SOS for the first time	Location and SMS permissions are requested	**Pass**
TC-SOS-02	Grant permissions and tap SOS	Live GPS/network location is obtained	**Pass**
TC-SOS-03	Check trusted-contact phone after SOS	SMS containing a Google Maps link is received	**Pass**
TC-SOS-04	Observe Firestore while SOS is active	Live location is updated about every 10 seconds	**Pass**
TC-SOS-05	Stay in an active SOS for several minutes	A fresh SMS is sent about every 2 minutes	**Pass**
TC-SOS-06	Check Nominee Dashboard and Admin Live SOS Monitor	Active SOS and location are displayed on both	**Pass**
TC-SOS-07	Tap I'm Safe	Safe message is sent, SOS stops, feedback/rating prompt opens	**Pass**
TC-SOS-08	Deny location or SMS permission	App handles the denial without crashing	**Pass**

7.4.5 Supporting Architecture

Table 7.6 Test Cases – Supporting Architecture

TC ID	Test Case / Steps	Expected Result	Status
TC-ARCH-01	Navigate through all main screens and use Back	Navigation is correct and consistent	**Pass**
TC-ARCH-02	Compare screens for colours, typography and shapes	Material 3 theme is applied consistently	**Pass**
TC-ARCH-03	Review code for Hilt injection and repository usage	Dependencies are injected; Firebase is accessed through repositories	**Pass**
TC-ARCH-04	Trigger dialogs and toasts from different screens	Reusable components behave consistently	**Pass**
TC-ARCH-05	Run SOS and check SOS Manager outputs	Location updates, SMS and synchronisation are handled by the SOS Manager	**Pass**

7.4.6 Intelligence & Notification

Table 7.7 Test Cases – Intelligence & Notification

TC ID	Test Case / Steps	Expected Result	Status
TC-INT-01	Ask the AI Assistant to explain a medicine	Relevant explanation is returned by the Gemini API	**Pass**
TC-INT-02	Ask a general health question	General guidance is returned	**Pass**
TC-INT-03	Use AI Assistant without internet	Error is handled without a crash	**Pass**
TC-INT-04	Open Alerts	Medicine reminders and recent activity are shown together	**Pass**
TC-INT-05	Wait for a scheduled medicine time	Notification is delivered at the scheduled time (AlarmManager)	**Pass**
TC-INT-06	Check notification with the app in the background	Notification is delivered	**Pass**

7.4.7 User & System

Table 7.8 Test Cases – User & System

TC ID	Test Case / Steps	Expected Result	Status
TC-USR-01	Open Profile and Logout	User information shown; logout returns to login	**Pass**
TC-USR-02	Edit profile and save	Updated information is shown and stored	**Pass**
TC-USR-03	Perform profile update, add nominee, trigger SOS; open Recent Activity	Each action is logged in the timeline	**Pass**
TC-USR-04	Toggle Dark/Light theme	Theme changes across the app	**Pass**
TC-USR-05	Open Help/User Manual, About, Privacy Policy, Terms of Service	Each page opens with its content	**Pass**
TC-USR-06	Submit feedback (1–5 stars + review)	Feedback is stored in Firestore and appears in the admin panel	**Pass**

7.4.8 Security

Table 7.9 Test Cases – Security

TC ID	Test Case / Steps	Expected Result	Status
TC-SEC-01	Enable PIN/biometric App Lock and reopen the app	Lock screen is shown before content	**Pass**
TC-SEC-02	Access Firestore data without authentication	Request is denied by security rules	**Pass**

TC ID	Test Case / Steps	Expected Result	Status
TC-SEC-03	User A attempts to read User B's data	Access is denied	**Pass**
TC-SEC-04	Nominee attempts to modify data	Write is not permitted (read-only)	**Pass**
TC-SEC-05	Check when location/SMS permissions are requested	Requested only during SOS	**Pass**
TC-SEC-06	Inspect network traffic of Firebase calls	Communication uses HTTPS	**Pass**
TC-SEC-07	Non-admin account accesses admin data	Access is denied	**Pass**

7.4.9 Web Admin Panel

Table 7.10 Test Cases – Web Admin Panel

TC ID	Test Case / Steps	Expected Result	Status
TC-WEB-01	Admin login with valid credentials	Admin dashboard opens	**Pass**
TC-WEB-02	Login with a non-admin account	Access is refused	**Pass**
TC-WEB-03	Compare dashboard figures with Firestore	Total users, active SOS, SOS alerts and average rating match the data	**Pass**
TC-WEB-04	Open analytics	Rating chart and 7-day SOS chart are displayed	**Pass**
TC-WEB-05	Open Live SOS Monitor during an active SOS	User and map location are shown	**Pass**
TC-WEB-06	Filter and search feedback	Ratings and reviews are filtered correctly	**Pass**
TC-WEB-07	Search Users, Contacts and Nominees pages	Searchable lists and counts are displayed	**Pass**
TC-WEB-08	Open the landing website	Animated page with features and team loads	**Pass**

7.4.10 System / End-to-End

Table 7.11 Test Cases – System / End-to-End

TC ID	Test Case / Steps	Expected Result	Status
TC-SYS-01	Full SOS scenario across owner phone, nominee device and admin panel	All parties see the SOS and location updates until I'm Safe	**Pass**
TC-SYS-02	Run on different Android versions and screen sizes	App installs and functions on each	**Pass**
TC-SYS-03	Create nominee with limited permissions, then trigger SOS	Nominee sees only permitted data plus the SOS alert	**Pass**

7.5 VALIDATION AGAINST OBJECTIVES

Table 7.12 Requirement Traceability

Objective / Requirement	Modules involved	Test case groups
Centralise emergency profile, medical records and documents	Core Data Management	TC-DATA
Granular, permission-based nominee access	Emergency & Access Control, Security	TC-ACC, TC-SEC
SOS emergency response with live location	Live SOS & Location, Supporting Architecture	TC-SOS, TC-ARCH, TC-SYS
Medicine reminders	Core Data Management, Intelligence & Notification	TC-DATA, TC-INT
AI-assisted health guidance	Intelligence & Notification	TC-INT
Secure sign-in and app protection	Authentication & Onboarding, Security	TC-AUTH, TC-SEC
Monitoring by administrators	Web Admin Panel, User & System	TC-WEB, TC-USR

7.6 QUALITY ASSURANCE CHECKLIST

Table 7.13 Quality Assurance Checklist

QA item	Method	Status
Firestore security rules reviewed for user, nominee and admin access	Rules review / emulator test	**Pass**
No API keys or secrets exposed in the repository or web files	Code review	**Pass**
Runtime permissions requested only when needed	Manual test	**Pass**
Consistent UI behaviour in light and dark themes	Manual test	**Pass**
Error handling for no network / permission denial	Manual test	**Pass**
AI Assistant clearly presented as general guidance	UI review	**Pass**
User manual and privacy policy match actual behaviour	Document review	**Pass**

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

LifeVault unifies emergency data, documents, medical records, and nominee access in one secure cloud application. The one-tap SOS shares the live location by SMS with trusted contacts and nominees until the user marks themselves safe, and the web admin panel monitors users, live SOS alerts, and feedback in real time.

By combining section-wise nominee permissions, biometric/Google authentication, AI-assisted health guidance, medicine reminders and digital legacy records, the project addresses the gaps identified in existing solutions and provides a single, connected platform for emergency preparedness. The extent to which the objectives were met is evidenced by the testing described in Chapter 7.

8.2 FUTURE SCOPE

The following enhancements are identified for future work:

- **Background live tracking** so that location continues to be shared when the application is not in the foreground.
- **Push notifications for nominees** so they are alerted immediately when an SOS starts.
- **Automatic fall detection** to trigger SOS without user action.
- **iOS version** of the application.
- **Hospital integration** for sharing emergency information with healthcare providers.
- **Advanced admin analytics** for deeper insight into usage and SOS activity.

CHAPTER 9

APPENDICES

APPENDIX A – SOURCE CODE

Web links: Landing/Admin website – https://lifevaultadmin123.netlify.app

1. SOS Flow

```
 fun triggerSos() {
        val locationOk = granted(Manifest.permission.ACCESS_FINE_LOCATION) ||
            granted(Manifest.permission.ACCESS_COARSE_LOCATION)
        val smsOk = granted(Manifest.permission.SEND_SMS)
        if (locationOk && smsOk) {
            viewModel.start(true, true)
        } else {
            permissionLauncher.launch(
                arrayOf(
                    Manifest.permission.ACCESS_FINE_LOCATION,
                    Manifest.permission.ACCESS_COARSE_LOCATION,
                    Manifest.permission.SEND_SMS
                )
            )
```

2.Nominee Permission Check

```
val permissions = (nomineeDoc.get("permissions")
    as? List<String>)?.toSet() ?: emptySet()
if ("EMERGENCY_PROFILE" in permissions)
if ("MEDICAL_RECORDS" in permissions)
if ("IMPORTANT_DOCUMENTS" in permissions)
```

3. Firestore Security Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    function isAdmin() {
```

```
    return request.auth != null && request.auth.token.email == 'admin@123lifevault.com';
  }
  match /users/{userId} {
    allow get: if true;
    allow list: if isAdmin();
  match /users/{userId}/{document=**} {
    allow read, write: if request.auth != null && request.auth.uid == userId;
```

4. Medicine Alarm Scheduling

```
val alarmManager =
  context.getSystemService(Context.ALARM_SERVICE) as AlarmManager
alarmManager.setRepeating(
  AlarmManager.RTC_WAKEUP,
  calendar.timeInMillis,
  AlarmManager.INTERVAL_DAY,
  pendingIntent
```

5. Gemini API Request/Response

```
val request = Request.Builder()
  .url("ENDPOINT?key =apiKey")
  .post(body.toRequestBody("application/json".toMediaType()))
  .build()

client.newCall(request).execute().use { response ->
  val responseBody = response.body?.string() ?: ""

  if (!response.isSuccessful) {
    return@withContext Result.failure(
      Exception("Gemini API error: $responseBody")
    )
  }

  val json = JSONObject(responseBody)
  val text = json.getJSONArray("candidates")
    .getJSONObject(0)
    .getJSONObject("content")
    .getJSONArray("parts")
    .getJSONObject(0)
    .getString("text")
}
```

APPENDIX B – SCREENSHOTS

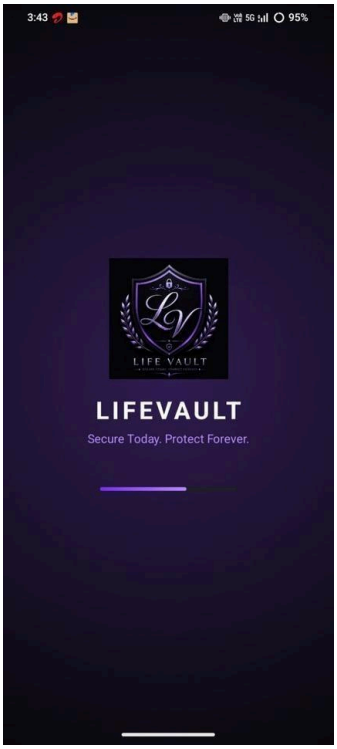


Fig.B.1.1 Splash Screen

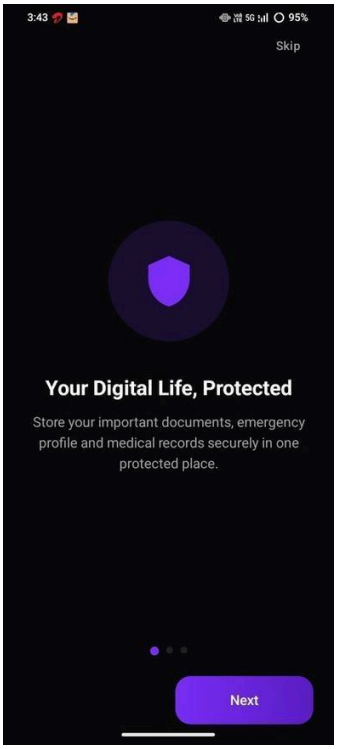


Fig.B.1.2 Onboarding Screen 1

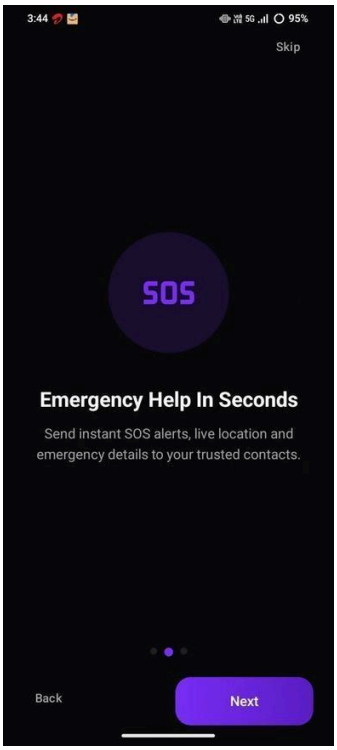


Fig.B.1.3 Onboarding Screen 2

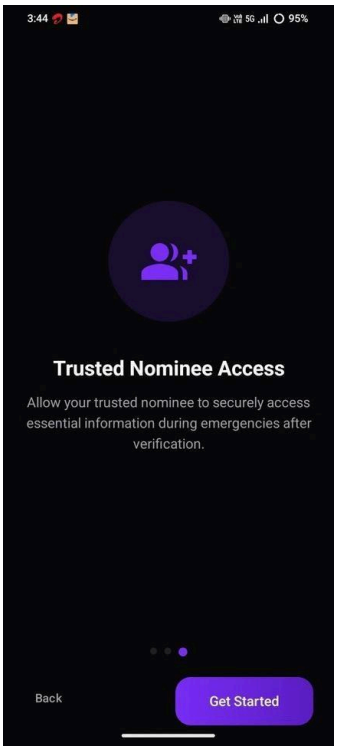


Fig.B.1.4 Onboarding Screen 3

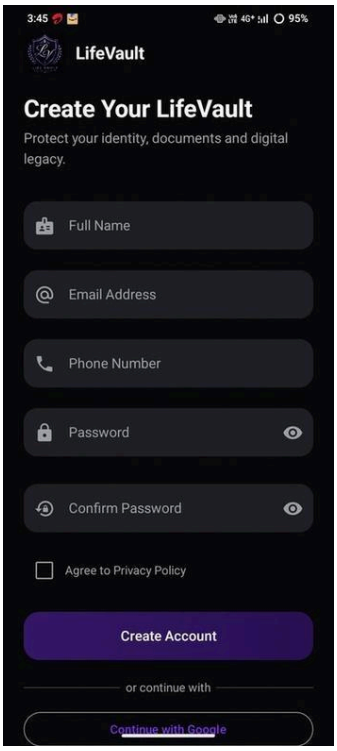


Fig.B.1.5 Register Page

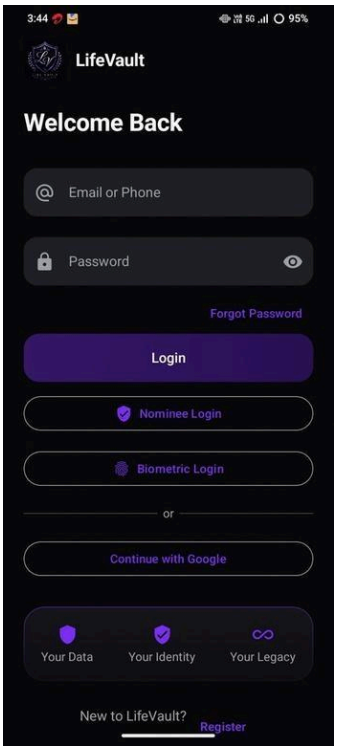


Fig.B.1.6 Login Page

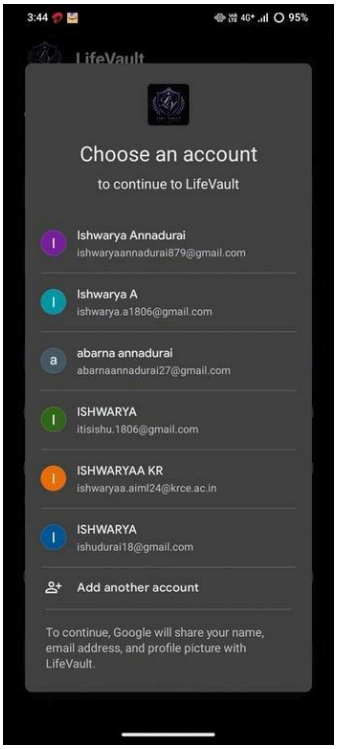


Fig.B.1.7 Google Sign-In Page

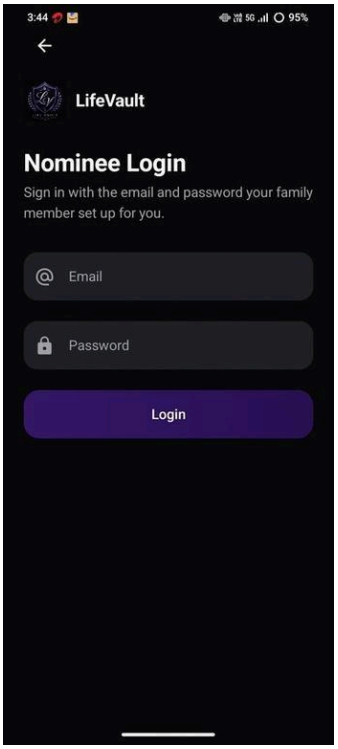


Fig.B.1.8 Nominee Login Page

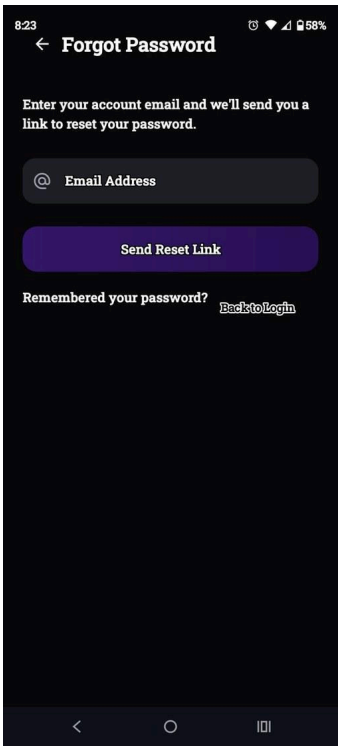


Fig.B.1.9 Forgot Password Page

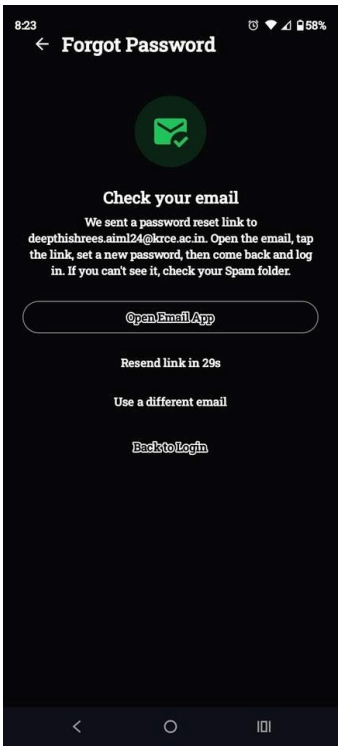


Fig.B.1.10 Forgot Password Navigation Page

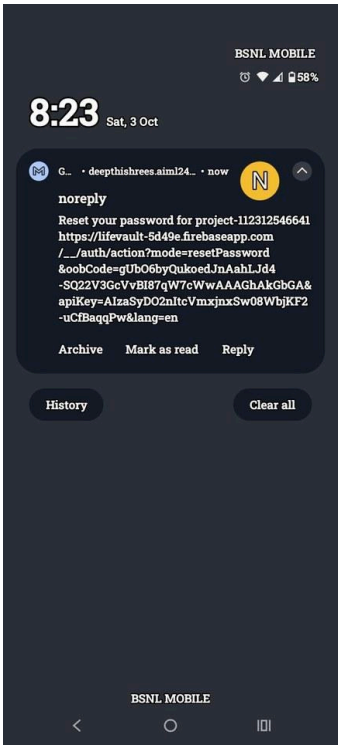


Fig.B.1.11 Forgot Password Reset Link Notification

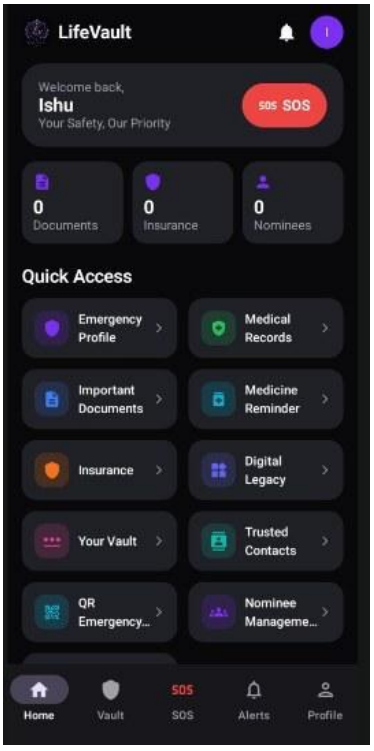


Fig.B.1.12 Dashboard Page

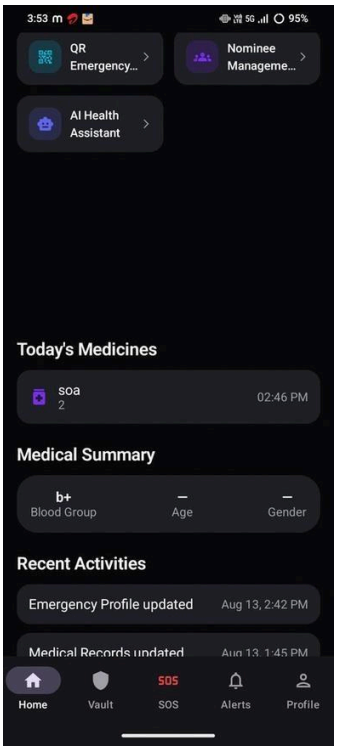


Fig.B.1.13 Dashboard and Recent Activity Page

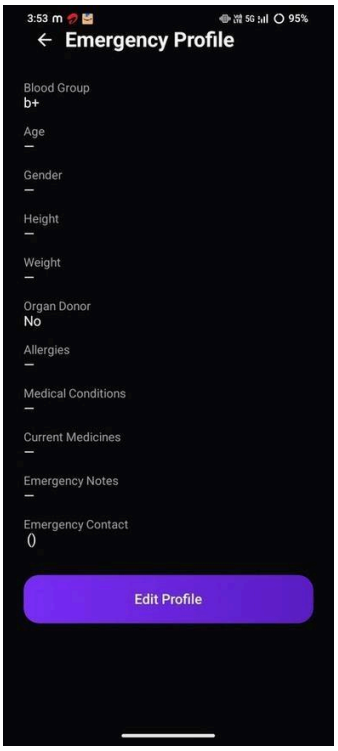


Fig.B.1.14 Emergency Profile Page

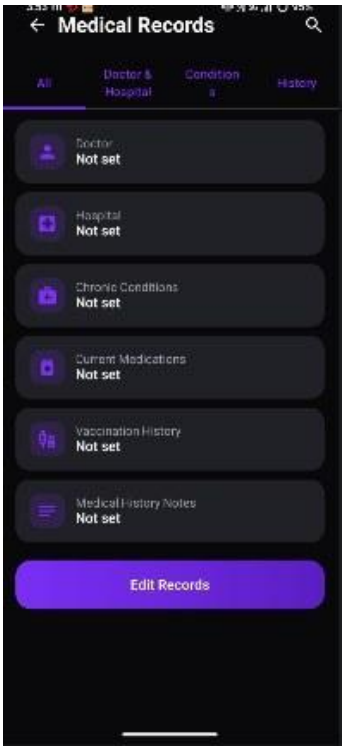


Fig.B.1.15 Medical Record Page

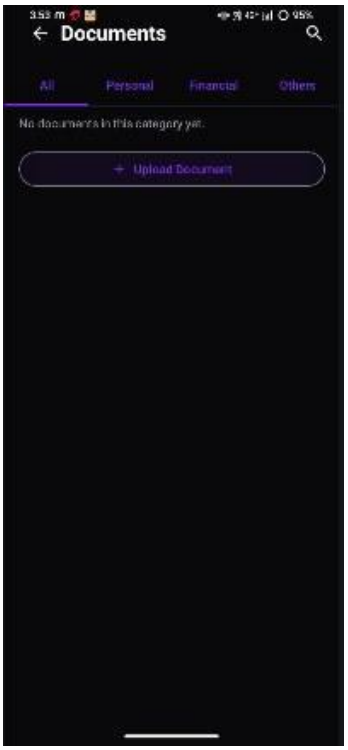


Fig.B.1.16 Important Document Page



Fig.B.1.17 Medicine Reminder Page

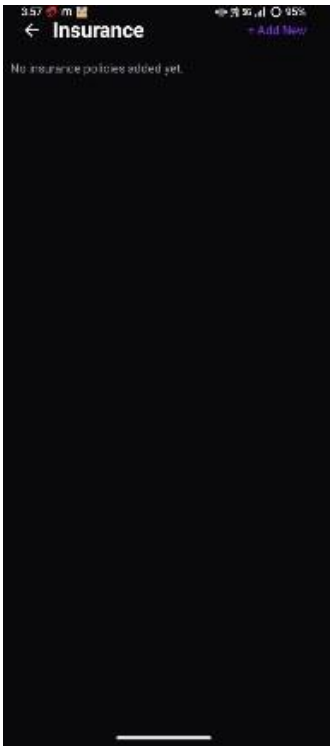


Fig.B.1.18 Insurance Detail Page

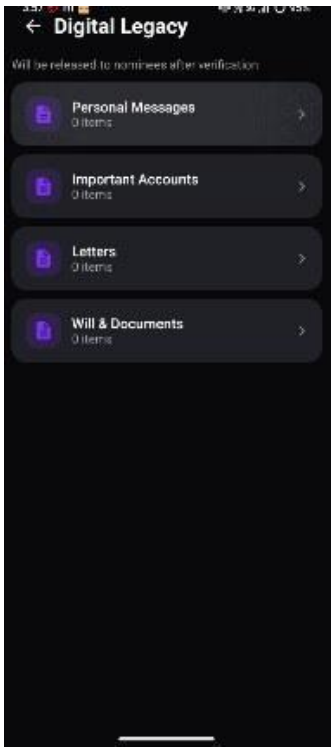


Fig.B.1.19 Digital Legacy Page

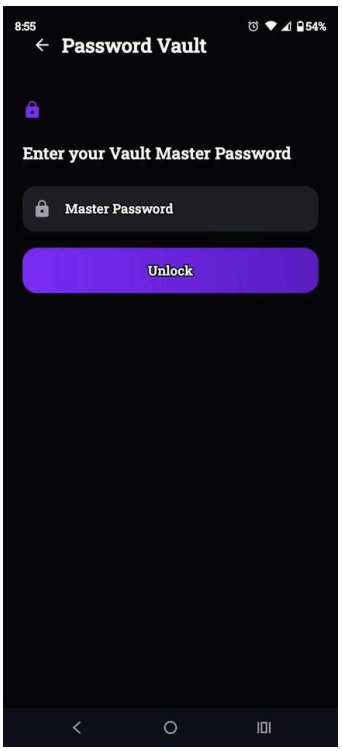


Fig.B.1.20 Password Vault Page



Fig.B.1.21 Passwords in Vault Page

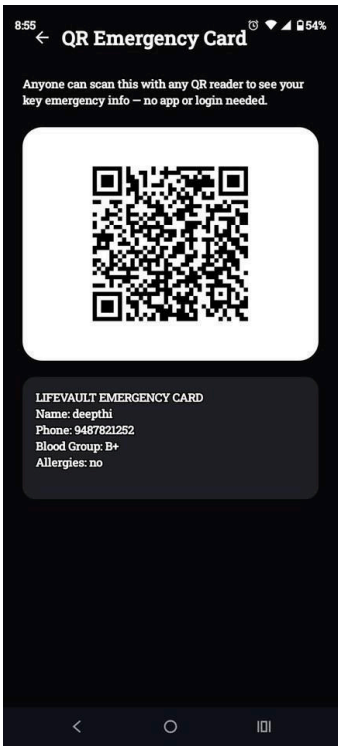


Fig.B.1.22 QR Emergency Card Page

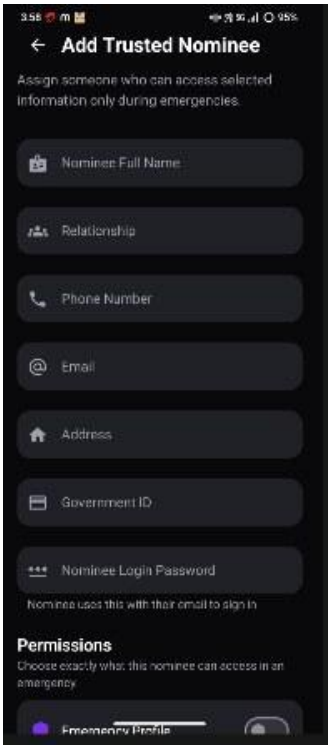


Fig.B.1.23 Nominee Creation Page (1)

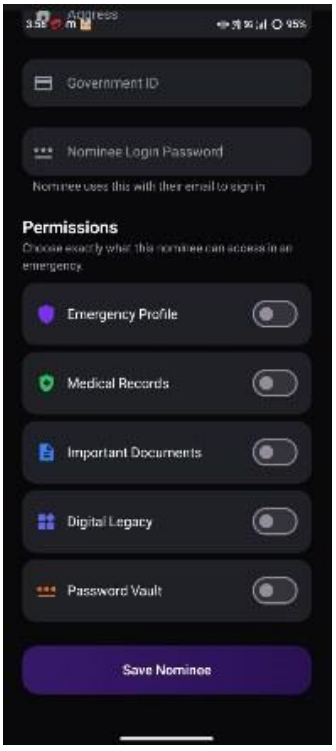


Fig.B.1.24 Nominee Creation Page (2)

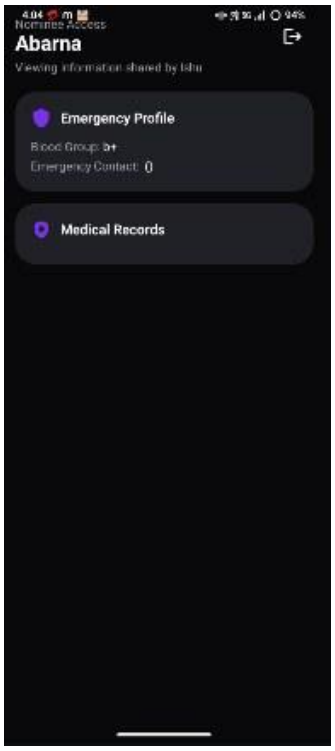


Fig.B.1.25 Nominee Dashboard Page

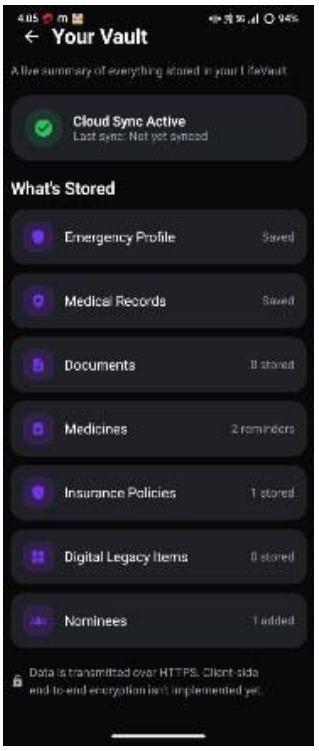


Fig.B.1.26 Storage Summary Page

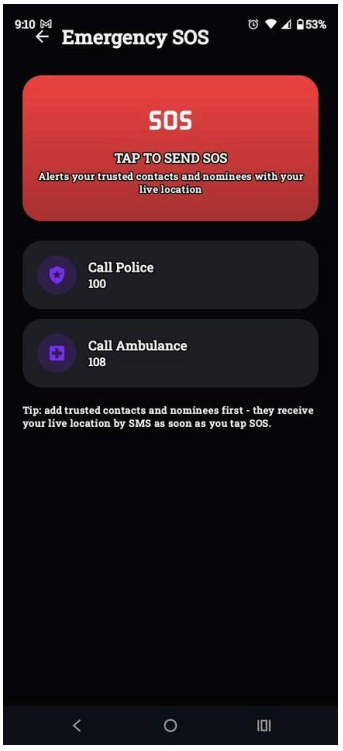


Fig.B.1.27 Emergency SOS Page



Fig.B.1.28 Call Police Page



Fig.B.1.29 Call Ambulance Page

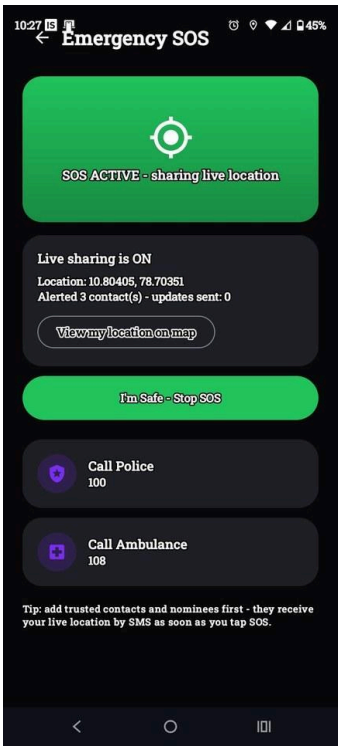


Fig.B.1.30 Live Location Send Page

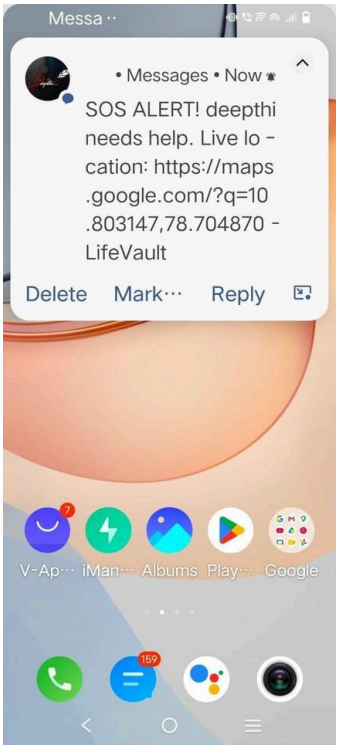


Fig.B.1.31 Live Location Notification

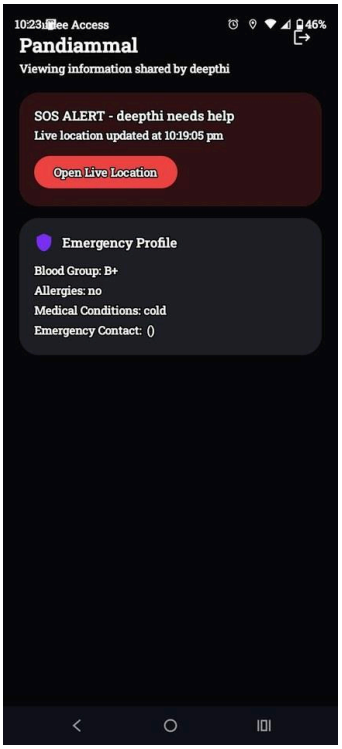


Fig.B.1.32 Live Location in Nominee Dashboard

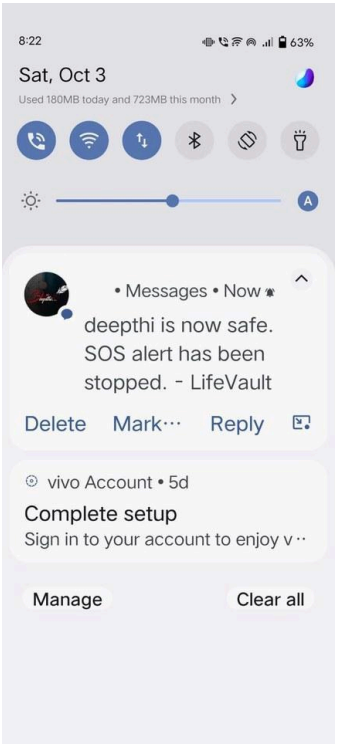


Fig.B.1.33 Live Location Stopped

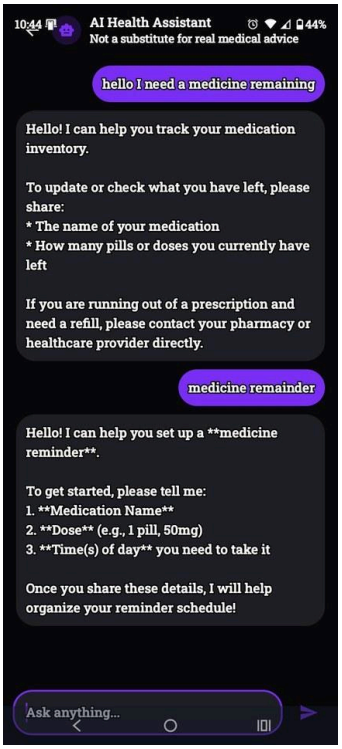


Fig.B.1.34 AI Assistant Chatbot

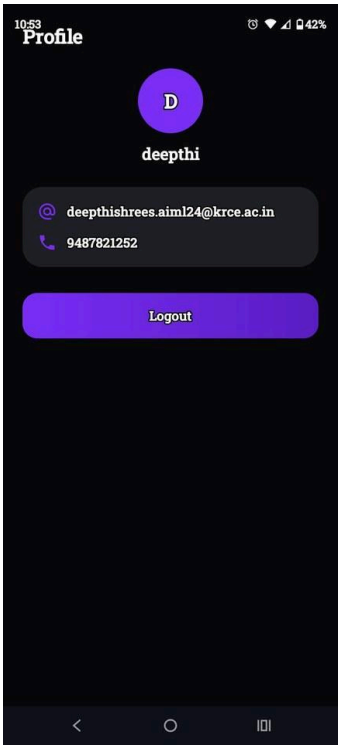


Fig.B.1.35 Profile Page

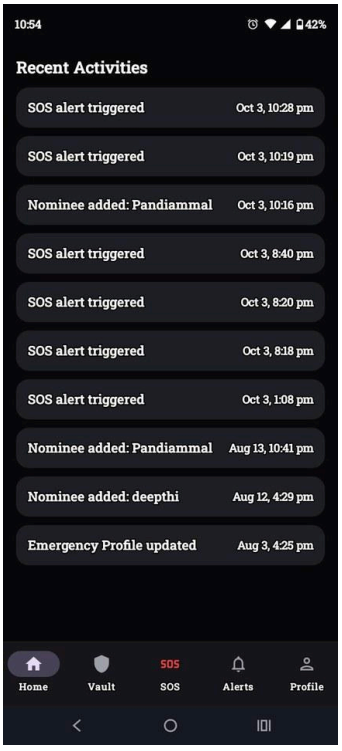


Fig.B.1.36 Recent Activity Page

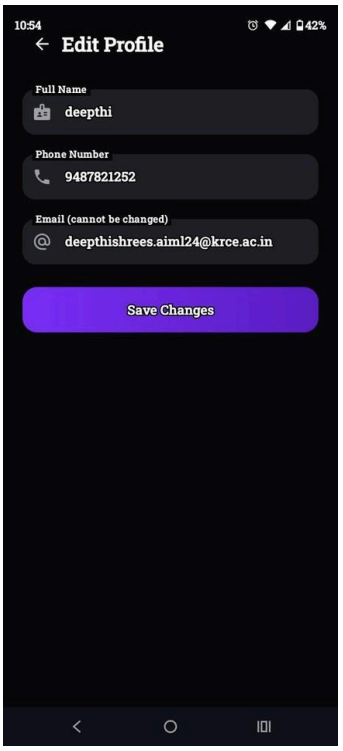


Fig.B.1.37 Edit Profile Page

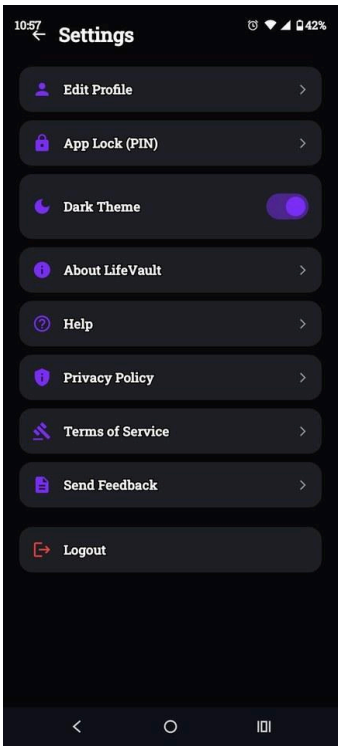


Fig.B.1.38 Dark / Light Theme Toggle Page

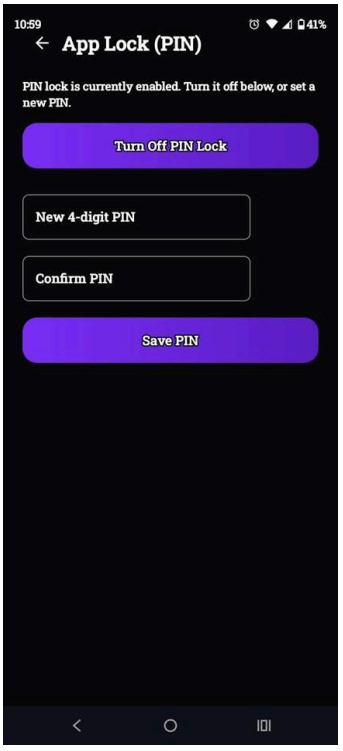


Fig.B.1.39 App Lock Page

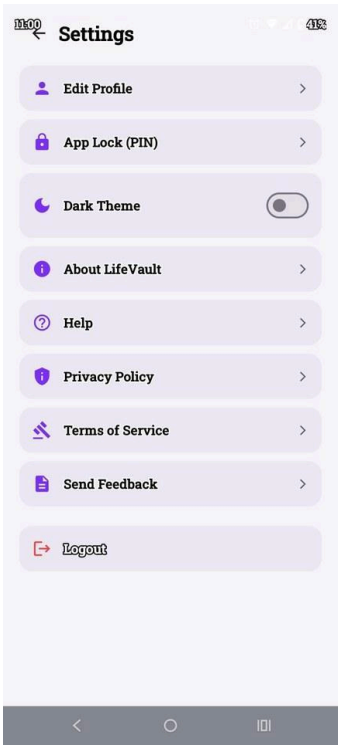


Fig.B.1.40 Settings Page

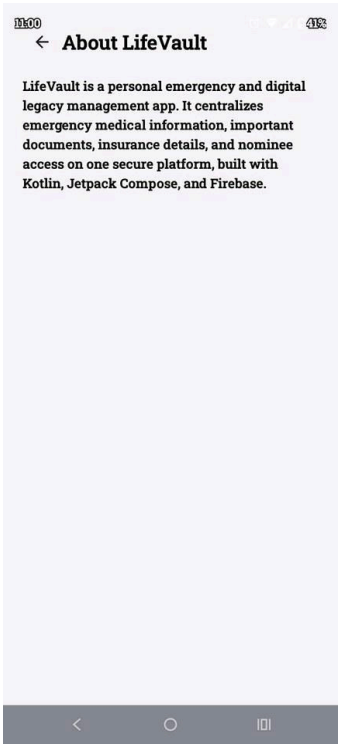


Fig.B.1.41 About Page

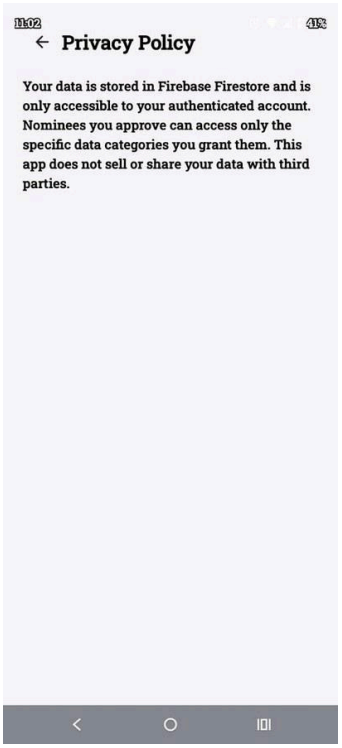


Fig.B.1.42 Privacy Policy Page

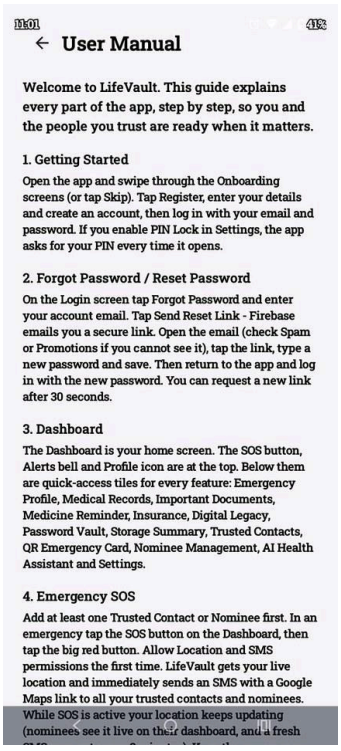


Fig.B.1.43 User Manual Page (1)

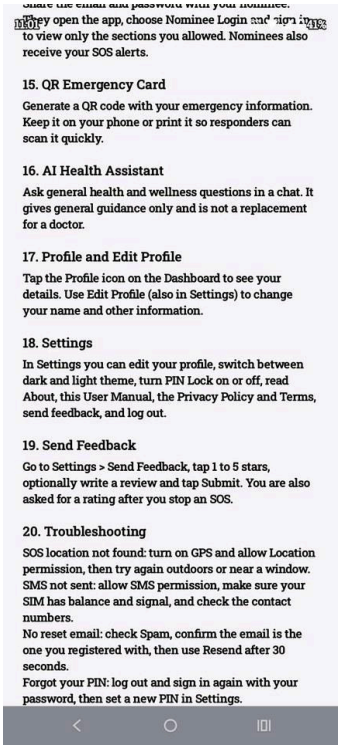


Fig.B.1.44 User Manual Page (2)

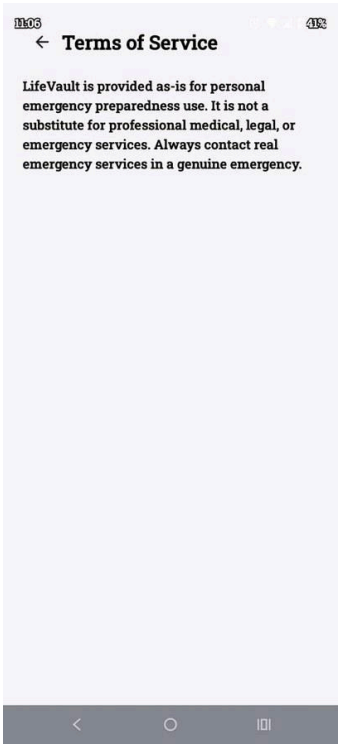


Fig.B.1.45 Terms of Service Page

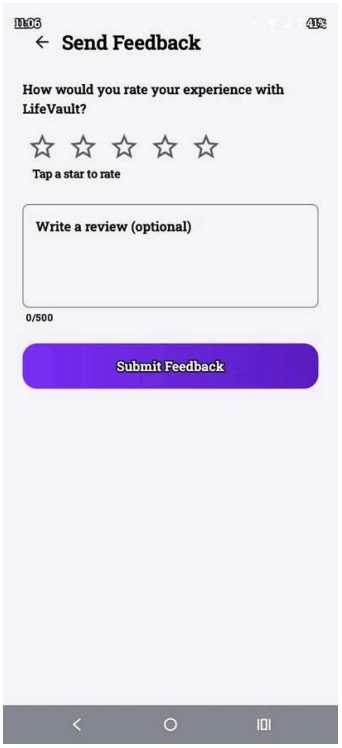


Fig.B.1.46 Feedback & Review Page

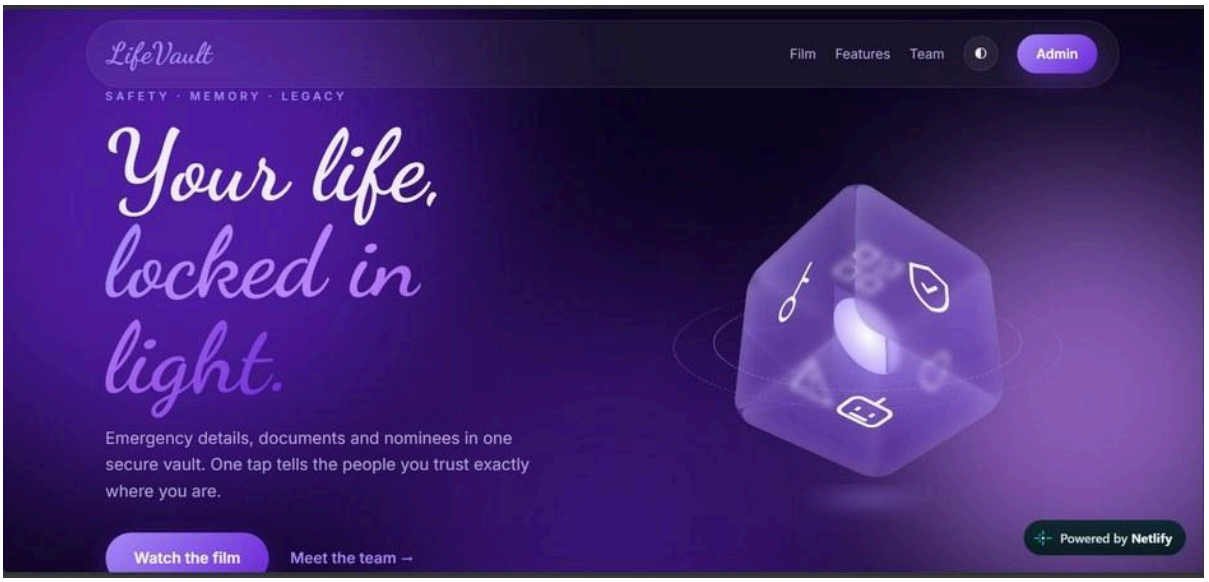


Fig.B.1.47 Landing Website

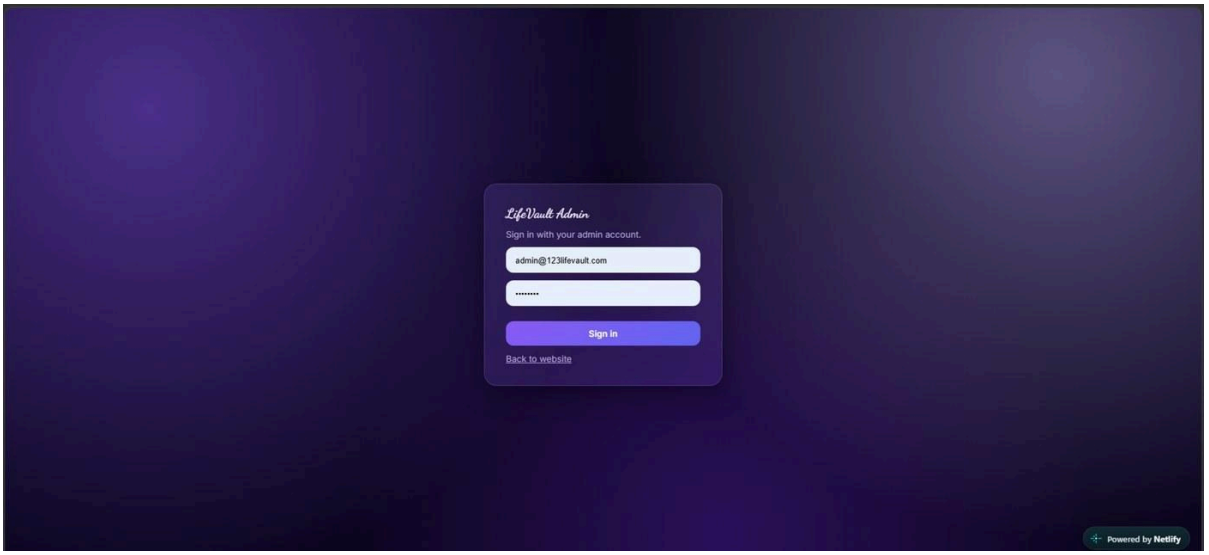


Fig.B.1.48 Admin Login Page

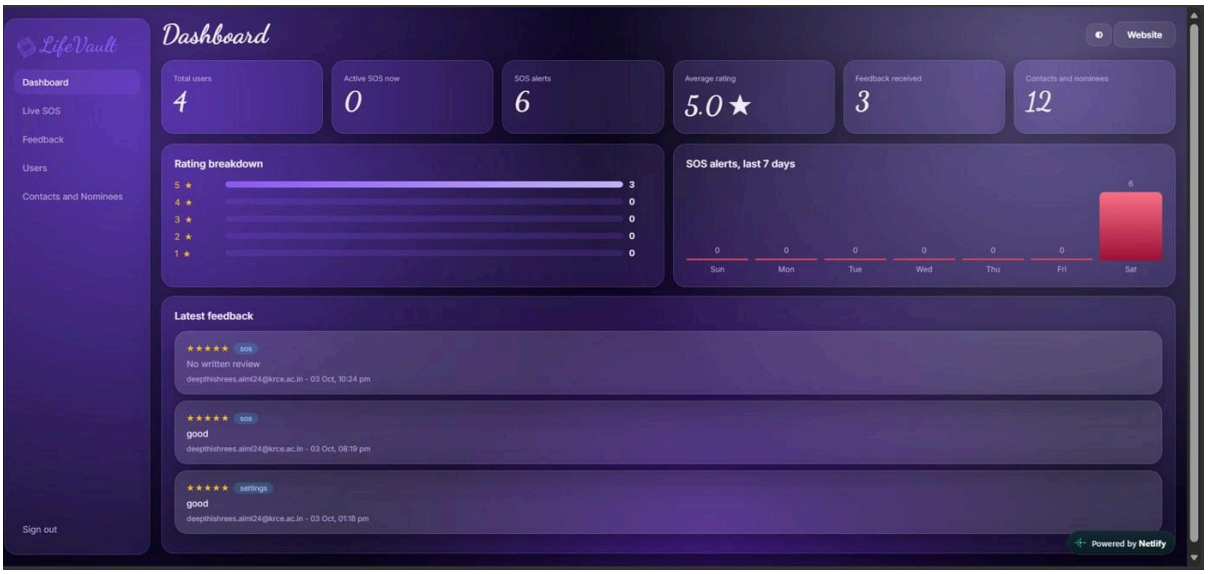


Fig.B.1.49 Admin Dashboard with 7-Day SOS Chart

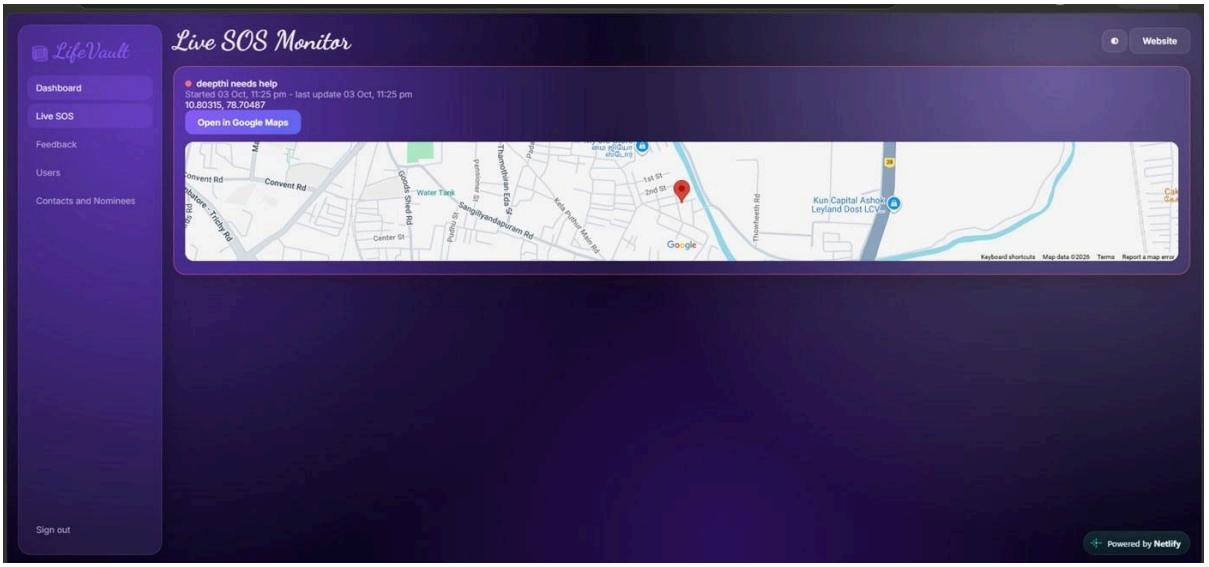


Fig.B.1.50 Live SOS Monitor + Map Page

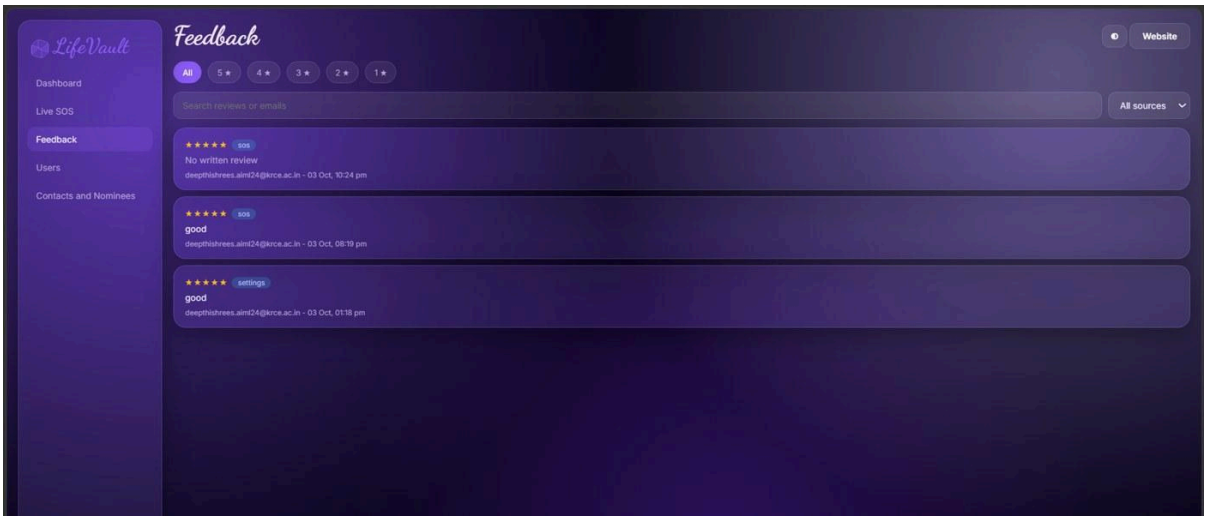


Fig.B.1.51 Feedback Management Page

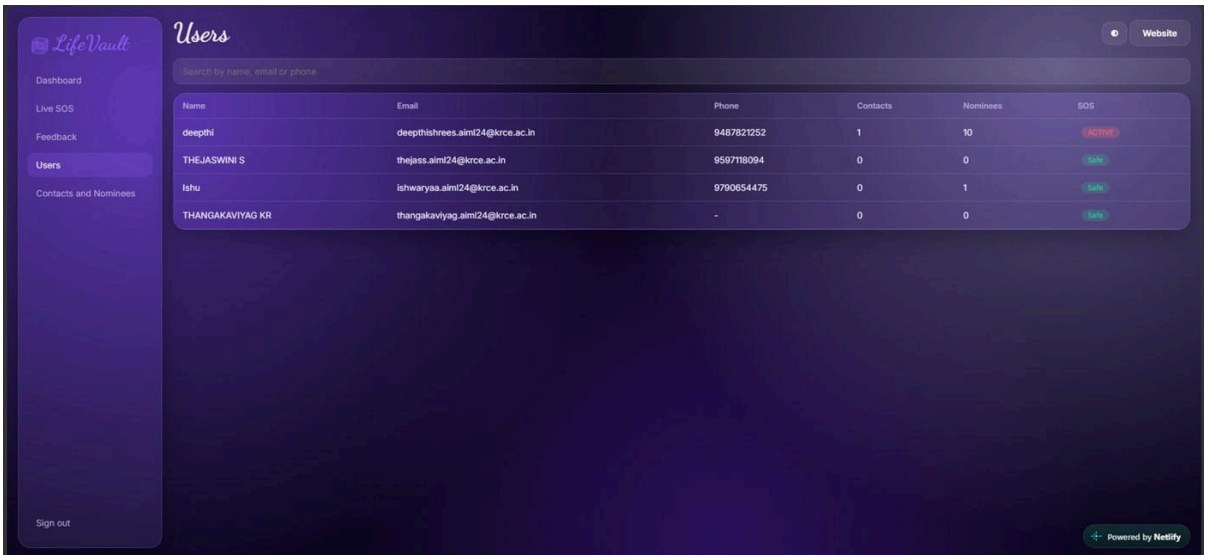


Fig.B.1.52 Users Management Page

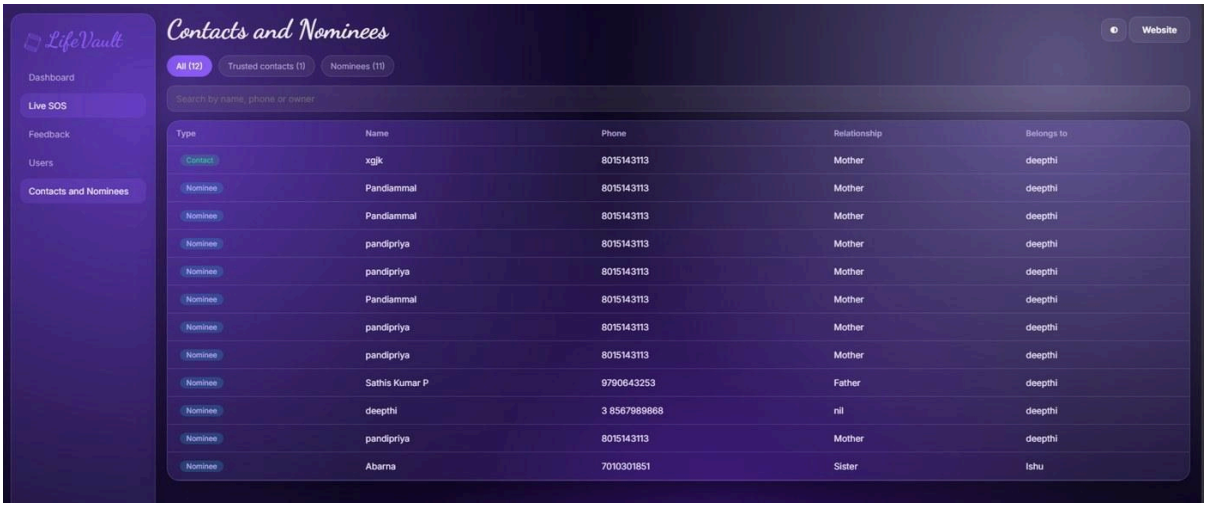


Fig.B.1.53 Nominee Management Page

CHAPTER 10
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